

A
Mini Project Report
on
Crypto Risk Lens

Submitted in partial fulfillment of the requirements for the
degree

**Third Year Engineering – Computer Science and Engineering
(Data Science)**

by

ADITYA KATE	23107126
SUMAN MANIK	23107056
TANMAY HARMALKAR	23107099

Under the guidance of
Prof. Rajashri Chaudhari



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(DATA SCIENCE)**

A.P. SHAH INSTITUTE OF TECHNOLOGY
G.B. Road, Kasarvadavali, Thane (W)-400615
UNIVERSITY OF MUMBAI

Academic year: 2025-26

CERTIFICATE

This to certify that the Mini Project report on **Crypto Risk Lens** has been submitted by **Aditya Kate (23107126)**, **Suman Manik (23107056)**, **Tanmay Harmalkar (23107099)** who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **Computer Science Engineering (Data Science)**, during the academic year **2025-2026** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

Ms. Rajashri Chaudhari
Guide

Dr. Pravin Adivarekar
HOD, CSE (Data Science)

Dr. Uttam D. Kolekar
Principal

External Examiner:
1.

Internal Examiner:
1.

Place: A. P. Shah Institute of Technology, Thane
Date:

ACKNOWLEDGEMENT

This project would not have come to fruition without the invaluable help of our guide **Ms. Rajashri Chaudhari** Expressing gratitude towards our HoD, **Dr. Pravin Adivarekar**, and the Department of Computer Science Engineering (Data Science) for providing us with the opportunity as well as the support required to pursue this project. We would also like to thank our Project coordinator **Ms. Sarala Mary** and **Ms. Priyanka Karhade** for their valuable suggestions and ideas when we needed them. Additionally, we extend our thanks to our peers for their helpful suggestions.

TABLE OF CONTENTS

Abstract

1. Introduction.....	1
1.1.Purpose.....	2
1.2.Problem Statement.....	3
1.3.Objectives.....	4
1.4.Scope.....	5-6
2. Literature Review.....	7-9
3. Proposed System.....	10
3.1. Features and Functionality.....	11-12
4. Requirements Analysis.....	13-14
5. Project Design.....	15
5.1.Use Case diagram.....	15-16
5.2.DFD (Data Flow Diagram)	17-19
5.3.System Architecture.....	20-21
5.4.Implementation.....	22-33
6. Technical Specification.....	34-37
7. Project Scheduling.....	38-39
8. Results.....	40-51
9. Conclusion.....	52
10. Future Scope.....	53

References

ABSTRACT

Cryptocurrency markets exhibit high volatility and complex behavior, making accurate risk assessment challenging. This paper presents Crypto Risk Lens, a real-time cryptocurrency risk analysis system integrating multiple machine learning techniques. XGBoost performs risk classification, Hidden Markov Models detect market regimes, K-Means clustering identifies behavioral patterns, and quantile regression forecasts volatility. System processes historical and real-time data including price movements, trading volume, and technical indicators. Sentiment analysis using Natural Language Processing extracts qualitative signals from news articles and social media, capturing market psychology alongside quantitative factors. Assets are classified into low, medium, and high-risk levels based on combined inputs. Modular architecture enables efficient data processing and interactive visualization through dashboard interface. XGBoost model achieves weighted F1 score of 0.652 with balanced precision and recall, demonstrating stable performance under volatile market conditions. Crypto Risk Lens offers unified, practical solution for real-time cryptocurrency risk monitoring by combining machine learning, sentiment analysis, and visualization into single cohesive platform.

Keywords - *Cryptocurrency Risk Analysis, XGBoost, Hidden Markov Models, Market Regime Detection, K-Means Clustering, Quantile Regression, Volatility Forecasting, Sentiment Analysis, Natural Language Processing, Risk Classification, Technical Indicators, Real-Time Analysis*

Chapter 1

Introduction

Cryptocurrency markets have emerged as a rapidly growing segment of the global financial ecosystem, offering new opportunities for investment and digital transactions. However, these markets are highly volatile and influenced by multiple dynamic factors such as price fluctuations, trading volume, market trends, and public sentiment. Unlike traditional financial markets, cryptocurrency prices can change drastically within short time intervals, often driven by news events, social media discussions, and investor behavior. This high level of uncertainty poses significant challenges for investors, analysts, and researchers in understanding and managing risk effectively. Existing tools and platforms primarily focus on historical price analysis and technical indicators, often neglecting the combined impact of sentiment and market behavior. As a result, users lack a comprehensive, real-time understanding of market risk and underlying trends.

To address these limitations, the Crypto Risk Lens project has been developed as an intelligent, real-time risk assessment system for cryptocurrency markets. The system integrates machine learning techniques with sentiment analysis to provide a unified view of market dynamics. It utilizes algorithms such as Random Forest and XGBoost for risk classification, K-Means clustering for identifying market patterns, and Hidden Markov Models for detecting market regimes such as stable, transitional, or high-volatility phases. Additionally, Natural Language Processing is applied to extract sentiment signals from news and social media sources, enabling the system to capture the influence of public perception on market movements. The architecture is designed using a FastAPI-based backend for data processing and a React-based interactive dashboard for visualization. The platform presents real-time insights including risk levels, volatility trends, sentiment scores, and market behavior patterns in a user-friendly manner. By combining market data with sentiment-driven analysis, the system enhances the understanding of cryptocurrency risk and supports more informed and data-driven decision-making for users.

1.1 Purpose

The purpose of this project extends beyond simple prediction to developing a comprehensive framework for understanding and managing cryptocurrency market risk. By integrating real-time market data with sentiment analysis, the system provides a holistic view of market behavior that is often missing in traditional trading tools. This enables investors to better understand risk levels, avoid impulsive decisions driven by market hype, and respond more effectively to sudden market changes. Analysts and researchers can utilize the platform to study patterns in volatility, sentiment shifts, and market regimes, leading to more informed and data-driven insights. Additionally, the system promotes responsible investment practices by presenting risk insights clearly, rather than encouraging speculative trading.

On a technological and analytical level, the project demonstrates the effective application of machine learning and data science techniques in financial risk analysis. Random Forest and XGBoost models enable accurate classification of cryptocurrency risk levels based on multiple technical indicators. K-Means clustering helps identify hidden market behavior patterns and group similar market conditions, while Hidden Markov Models capture underlying market regimes such as stable, transitional, and high-volatility phases. Furthermore, Natural Language Processing techniques are used to extract sentiment from news and social media, allowing the system to incorporate the impact of public perception into risk evaluation. This integration bridges the gap between numerical data analysis and real-world market sentiment.

Moreover, the project is designed with scalability and adaptability in mind. Although focused on cryptocurrency markets, the modular architecture allows it to be extended to other financial domains such as stock markets or forex trading. The combination of real-time data processing, machine learning models, and interactive visualization highlights how modern technologies can transform financial analysis into a more intelligent, transparent, and user-centric process. By unifying data analysis, sentiment understanding, and risk evaluation, Crypto Risk Lens contributes to building a more informed and stable digital financial ecosystem.

1.2 Problem Statement

Cryptocurrency markets are known for their extreme volatility and rapid fluctuations, often influenced by a combination of price movements, trading behavior, and external factors such as news and social media sentiment. Sudden price crashes, market manipulation, and hype-driven trends make it difficult for investors to accurately assess risk and make informed decisions. Unlike traditional financial systems, the decentralized and highly dynamic nature of crypto markets increases uncertainty and exposes users to higher financial risks. These challenges highlight the need for a more reliable and intelligent system to monitor and interpret market behavior in real time.

1. Limitations of Current Systems

Most existing cryptocurrency analysis tools focus primarily on historical price data and technical indicators. While these tools provide useful insights, they often ignore the impact of real-time sentiment and external information sources. Additionally, many platforms do not offer integrated analysis that combines risk prediction, volatility forecasting, sentiment evaluation, and market regime detection. As a result, users receive fragmented information, making it difficult to understand the complete risk landscape.

2. Challenges for Users and Analysts

Investors and analysts lack access to unified, real-time risk assessment systems that can interpret both quantitative market data and qualitative sentiment signals. This limitation leads to delayed or uninformed decision-making, where users may react to market changes only after significant losses have occurred. Furthermore, the absence of clear risk classification and explainable insights makes it difficult for beginners to navigate the complex crypto ecosystem, increasing the chances of speculative and high-risk investments.

3. Need for Advanced Solutions

There is a growing need for an intelligent system that integrates real-time market data with sentiment analysis to provide a comprehensive view of cryptocurrency risk. Such a system should utilize machine learning models to classify risk levels, detect market regimes, and forecast volatility while incorporating sentiment from news and social media. The platform should present these insights through an interactive and user-friendly interface, enabling users to make informed, data-driven decisions and better understand market dynamics.

1.3 Objectives

The main objective of this project is to design and implement a real-time cryptocurrency risk assessment and analysis system that provides comprehensive insights into market behavior and volatility. To achieve this goal, the system is guided by the following focused objectives:

1. Predict Flood Risk Using Random Forest and XGBoost

The primary objective is to classify cryptocurrency assets into risk categories such as low, medium, and high using machine learning models like Random Forest and XGBoost. These models are trained on historical and real-time market data, including price, volume, and technical indicators, to generate accurate and data-driven risk predictions.

2. Analyze Market Behavior Using Clustering

The second objective is to identify patterns in market behavior using clustering techniques such as K-Means. By grouping cryptocurrencies or time periods with similar characteristics, the system uncovers hidden structures and recurring trends, helping users understand different market conditions.

3. Detect Market Regimes Using Hidden Markov Models

The third objective is to detect underlying market regimes such as stable, transitional, and high-volatility phases using Hidden Markov Models. This helps in understanding shifts in market conditions and provides deeper insights into long-term market dynamics.

4. Integrate Sentiment Analysis Using NLP

The fourth objective is to analyze sentiment from real-time news and social media data using Natural Language Processing techniques. This allows the system to capture the impact of public perception and external events on cryptocurrency markets, enhancing the accuracy of risk assessment.

5. Provide Volatility Forecasting and Interactive Visualization

Estimate future market volatility using regression techniques such as Linear and Quantile Regression, and present all insights through an interactive dashboard. The platform visualizes risk levels, sentiment scores, market trends, and volatility using charts and graphs, enabling users to easily interpret complex data and make informed decisions.

1.4 Scope

The scope of this project encompasses the design, development, implementation, and evaluation of a comprehensive, AI-driven cryptocurrency risk assessment system. The project focuses on integrating real-time market data, technical indicators, and sentiment analysis to provide a unified and intelligent framework for analyzing cryptocurrency market behavior. It aims to overcome the limitations of existing tools that rely only on price-based analysis by delivering real-time, data-driven risk insights through an interactive and user-friendly platform.

1. Data integration and monitoring

The system integrates multiple data sources to ensure accurate and reliable analysis. Historical and real-time cryptocurrency market data, including price and volume (OHLCV), are collected from platforms like CoinGecko. In addition, sentiment data is gathered from news sources, RSS feeds, and social media platforms. All data undergo preprocessing and feature engineering, including the calculation of technical indicators such as RSI and MACD, to support machine learning models and real-time analysis.

2. Stakeholder Benefits

The project caters to a wide range of stakeholders. Investors and traders receive real-time risk insights and sentiment analysis to make informed decisions and avoid high-risk investments. Analysts and researchers benefit from deeper understanding of market behavior, volatility patterns, and regime shifts. Additionally, the platform promotes responsible investing by providing clear risk classifications rather than direct trading recommendations.

3. Limitations and Dependencies

The performance of the system depends on the availability, accuracy, and consistency of real-time market and sentiment data. Delays or failures in API services or data sources may affect prediction accuracy. Furthermore, cryptocurrency markets are highly unpredictable, and sudden events such as regulatory changes or market manipulation may not always align with historical patterns, limiting model reliability. Continuous internet connectivity and system uptime are also required for real-time analysis.

4. Scalability and Future Adaptation

Although currently focused on cryptocurrency markets, the system is designed with scalability in mind. Its modular architecture allows easy adaptation to other financial domains such as stock markets or forex trading. The integration of machine learning, sentiment analysis, and interactive visualization creates a flexible framework that can evolve with changing market conditions and technological advancements.

Chapter 2

Literature Review

This literature review examines existing research on cryptocurrency market analysis, risk prediction, and sentiment-based forecasting. The selected studies cover a wide range of approaches including machine learning models, deep learning techniques, sentiment analysis, and hybrid frameworks. By analyzing these contributions, the section highlights key findings, methodologies, and limitations in current crypto risk analysis systems. The aim is to identify research gaps and justify the need for an integrated, real-time risk assessment system combining market data and sentiment analysis.

Jutur Manogna et al. (2024) [1] proposed a cryptocurrency price prediction model that incorporates sentiment analysis derived from news and textual data. The methodology focuses on extracting sentiment scores and combining them with traditional machine learning models to improve prediction accuracy. The results indicated that sentiment plays a significant role in influencing price movements, especially during periods of market uncertainty. However, the study primarily focuses on price prediction rather than overall market risk, limiting its ability to provide comprehensive insights into volatility and long-term stability.

Andrew Coutinho et al. (2024) [2] developed a multi-platform sentiment analysis framework that collects data from various sources such as news websites, social media, and online discussions. The framework integrates sentiment signals with price data to enhance predictive performance. While this approach improves accuracy by capturing diverse data sources, it lacks integration with technical indicators and does not provide a structured risk classification system. This highlights the limitation of focusing only on sentiment without combining it with broader market analytics.

Vincent Oei et al. (2024) [3] investigated the effectiveness of combining sentiment analysis with Random Forest models for predicting Bitcoin prices. The study demonstrated that incorporating sentiment features significantly improves model performance compared to using only historical price data. However, the research is limited to a single cryptocurrency and does not consider market-wide behavior such as clustering of assets or identification of different market phases, reducing its applicability for comprehensive risk analysis.

Shalini Singh et al. (2024) [4] explored risk analysis in cryptocurrency markets with a focus on stablecoins and their impact on financial systems. The study emphasized the importance of understanding risk factors such as market liquidity, price stability, and external economic influences. While it provides valuable insights into financial risk, the approach lacks real-time predictive capabilities and does not incorporate machine learning models for dynamic risk assessment.

Sumin Li et al. (2025) [5] applied advanced deep learning models such as LSTM and FEDformer for cryptocurrency price prediction, enhanced with sentiment analysis. These models are effective in capturing temporal dependencies and complex patterns in time-series data. However, LSTM-based approaches are computationally intensive, require large datasets, and often suffer from overfitting in highly volatile markets. Additionally, their black-box nature reduces interpretability, making them less suitable for systems that require transparent and explainable risk assessment.

Saeed Mohammadi Dashtaki et al. (2025) [6] proposed a transformer-based framework using cross-attention mechanisms to combine market and sentiment data for trend prediction. This approach effectively models complex relationships between multiple data sources and improves forecasting performance. However, transformer-based models require high computational resources and are difficult to deploy in real-time systems with limited infrastructure, posing challenges for scalability and practical implementation.

Yahia et al. (2024) [7] proposed a machine learning-based approach for anomaly detection in cryptocurrency markets using unsupervised learning techniques such as Local Outlier Factor (LOF), Isolation Forest, and One-Class Support Vector Machine (SVM). The study focuses on identifying abnormal patterns in cryptocurrency return data to detect irregular market behavior, sudden price fluctuations, or potential manipulation. Among the evaluated models, LOF demonstrated better performance in capturing local anomalies and detecting deviations in market trends. However, the approach is limited to anomaly detection and does not provide a complete risk classification or prediction framework, and it lacks integration with real-time data processing and user-oriented systems.

Palaiokrassas et al. (2024) [8] proposed a machine learning-based framework for credit risk assessment and liquidation prediction in decentralized finance (DeFi) systems. The study utilizes various financial and transactional features from blockchain data to predict the likelihood of wallet liquidation using classification models evaluated through metrics such as

accuracy, precision, recall, and AUC. The approach demonstrates strong predictive capability in identifying high-risk lending positions and improving financial risk management in DeFi platforms. However, the system is limited to a specific domain of DeFi and does not address broader cryptocurrency market risk, and it is primarily evaluated in an experimental setting without real-time deployment or integration with interactive systems.

System / Paper	F1 Score	Model Type	Task Complexity	Remarks
Price Prediction Bitcoin (CNN-BiLSTM)	0.98	Deep Learning (CNN + BiLSTM)	Low (Binary Classification)	Performs sentiment classification (positive/negative). High F1 due to simpler task and controlled dataset.
Bitcoin Price Prediction (CNN-LSTM)	0.97	Deep Learning (CNN + LSTM)	Low (Binary Classification)	Focuses on sentiment-based prediction. High accuracy achieved due to limited class complexity.
HSIF Transformer Model	0.9727	Transformer (FinBERT-based)	Medium (Trend Classification)	Combines sentiment and market data. High performance but evaluated in offline experimental setup.
Crypto Risk Lens (Proposed)	0.652	Hybrid ML System (XGBoost + HMM + Clustering)	High (Multi-class + Real-Time)	Performs real-time multi-class risk classification using noisy data, making it more complex and practical.

Table 2.1. Literature Survey Summary on F1 Score Evaluation

The table 2.1 presents a comparison of classification-based models using the F1 score as a common evaluation metric. It is observed that existing models achieve very high F1 scores due to simpler tasks such as binary classification and evaluation on controlled datasets. In contrast, the proposed Crypto Risk Lens system performs multi-class risk classification on real-time, noisy data, making the problem more complex. Therefore, although the F1 score is comparatively lower, it reflects a more practical and realistic system performance.

Chapter 3

Proposed System

The proposed system, Crypto Risk Lens, aims to enhance cryptocurrency market analysis by providing accurate, real-time risk assessment and sentiment insights across multiple digital assets. By integrating machine learning with interactive visualizations and real-time data processing, the system enables investors, analysts, and researchers to better understand market volatility, behavior, and risk, supporting more informed and data-driven decision-making.

1. The System Architecture: Crypto Risk Lens follows a scalable multi-layer architecture, including a React-based frontend for interactive dashboards, a FastAPI backend for API management, machine learning modules for predictions, and a data layer for handling historical and real-time market inputs.

2. Frontend Interface: The web dashboard uses React and TypeScript to display real-time cryptocurrency data, risk levels, sentiment scores, and market trends through interactive charts and visualizations, allowing users to explore insights efficiently.

3. API Gateway: Implemented using FastAPI and Uvicorn, this layer manages RESTful endpoints for market data retrieval, risk prediction, and sentiment analysis, incorporating error handling, asynchronous processing, and efficient data communication.

4. Machine Learning Models: The system uses an ensemble approach combining Random Forest and XGBoost for risk classification, K-Means clustering for identifying market patterns, and Hidden Markov Models for detecting market regimes, trained on historical and real-time datasets.

5. Data Integration: The system integrates market data such as price, volume, and OHLCV from CoinGecko along with sentiment data from news sources and social media, applying preprocessing and feature engineering techniques to generate meaningful inputs for analysis.

6. Analysis and Visualization: Natural Language Processing techniques extract sentiment signals, which are combined with technical indicators to generate risk insights, presented through an interactive dashboard with charts and graphs for easy interpretation.

By implementing these features, Crypto Risk Lens aims to provide a comprehensive platform for real-time cryptocurrency risk analysis, improving market understanding and supporting effective decision-making in a highly volatile environment.

3.1 Features and Functionality

Real-Time Risk Assessment:

The system provides real-time cryptocurrency risk evaluation using a combination of live market data and historical patterns. Users can view risk levels for individual cryptocurrencies through an interactive dashboard, with assets categorized into risk levels such as low, medium, and high based on machine learning model predictions.

Interactive Data Visualization:

The dashboard enables users to explore market data through an intuitive interface. It displays charts for price trends, volatility, and risk levels, along with sentiment scores. Users can interact with graphs, switch between assets, and analyze historical and real-time data effectively.

Sentiment Analysis System:

The platform integrates sentiment analysis by extracting data from news sources and social media. Natural Language Processing techniques are used to generate sentiment scores, helping users understand how public opinion and external events influence market behavior.

Market Behavior Analysis:

The system identifies patterns in cryptocurrency markets using clustering techniques such as K-Means. It groups assets or time periods with similar characteristics, enabling users to recognize trends and recurring market conditions.

Market Regime Detection:

Hidden Markov Models are used to detect different market regimes such as stable, transitional, and high-volatility phases. This helps users understand shifts in market conditions and anticipate potential risks.

Volatility Forecasting:

The system predicts future market volatility using regression models, providing insights into expected price fluctuations and uncertainty levels over time.

Batch Analysis Processing:

The platform supports analysis of multiple cryptocurrencies simultaneously, allowing users to monitor overall market trends and compare risk levels across different assets.

Model Performance Monitoring:

The system includes monitoring features to track model performance, prediction accuracy, and system health. Logs and metrics are maintained for continuous improvement and reliability.

Caching and Performance Optimization:

Efficient caching mechanisms ensure fast response times for data retrieval and predictions. The system uses optimized data handling techniques to support real-time analysis without delays.

Fallback and Error Handling:

Robust error handling and fallback mechanisms ensure system stability during API failures or data unavailability. The platform maintains basic functionality and provides alternative data handling strategies to ensure uninterrupted user experience.

Chapter 4

Requirements Analysis

For the requirement analysis of the Crypto Risk Lens project, it is essential to identify and document the functional and non-functional requirements that the system must satisfy to perform real-time cryptocurrency risk assessment effectively. The following outlines the key requirements of the system:

A. Functional Requirements:

1. Data Collection and Integration:

The system should collect real-time cryptocurrency market data such as price, volume, and OHLCV from APIs like CoinGecko, and integrate it with historical datasets. It should also gather sentiment data from news sources, RSS feeds, and social media platforms to ensure comprehensive data availability.

2. Data Preprocessing:

The system should preprocess collected data by handling missing values, removing inconsistencies, and performing feature engineering. It should compute technical indicators such as RSI, MACD, and volatility measures for accurate analysis.

3. Crypto Risk Prediction:

The system should analyze processed data using machine learning models such as Random Forest and XGBoost to classify cryptocurrencies into risk categories like low, medium, and high.

4. Market Behaviour Analysis:

The system should apply clustering techniques such as K-Means to identify patterns and group similar market conditions or assets for better understanding of market behavior

5. Market Sentiment Analysis:

The system should analyse textual data from news and social media using NLP techniques to generate sentiment scores and integrate them with market analysis.

6. Market Regime Detection:

The system should detect different market phases such as stable, transitional, and high-volatility conditions using models like Hidden Markov Models.

7. Visualization and Dashboards:

The system should display results through an interactive dashboard with charts, graphs and visual indicators representing risk levels, sentiment trends, volatility and market patterns.

B. Non-Functional Requirements:

1. Real-Time Performance:

The system should process and update data in real time to provide timely risk assessments and insights for rapidly changing cryptocurrency markets.

2. Accuracy:

The system should maintain high prediction accuracy to ensure reliable risk classification and meaningful insights for users.

3. Reliability and Robustness:

The system should handle large volumes of data and multiple concurrent users without system failure, ensuring consistent performance.

4. Usability:

The system should provide an intuitive and user-friendly interface, allowing users with different levels of technical knowledge to easily interpret market insights.

5. Software Infrastructure:

The system should utilize python for backend development, FastAPI for API handling, and machine learning libraries such as Scikit-learn and XGBoost. The frontend should be developed using React, TypeScript and visualization libraries like Recharts, D3 for efficient and interactive user experience.

Chapter 5

Project Design

This chapter presents system design and implementation of Crypto Risk Lens. It covers Use Case Diagram illustrating user interactions with system, Data Flow Diagram depicting flow of data across modules, System Architecture describing structural organization of machine learning components, and Implementation Screenshots demonstrating frontend interface of system.

5.1. Use Case diagram

The Use Case Diagram of Crypto Risk Lens (fig 5.1) captures the main functionalities and interactions between the user and the system. It illustrates how users interact with the cryptocurrency risk analysis platform and how the system processes data internally to generate meaningful insights. The diagram highlights the key operations involved in real-time market analysis, risk prediction, and sentiment evaluation.

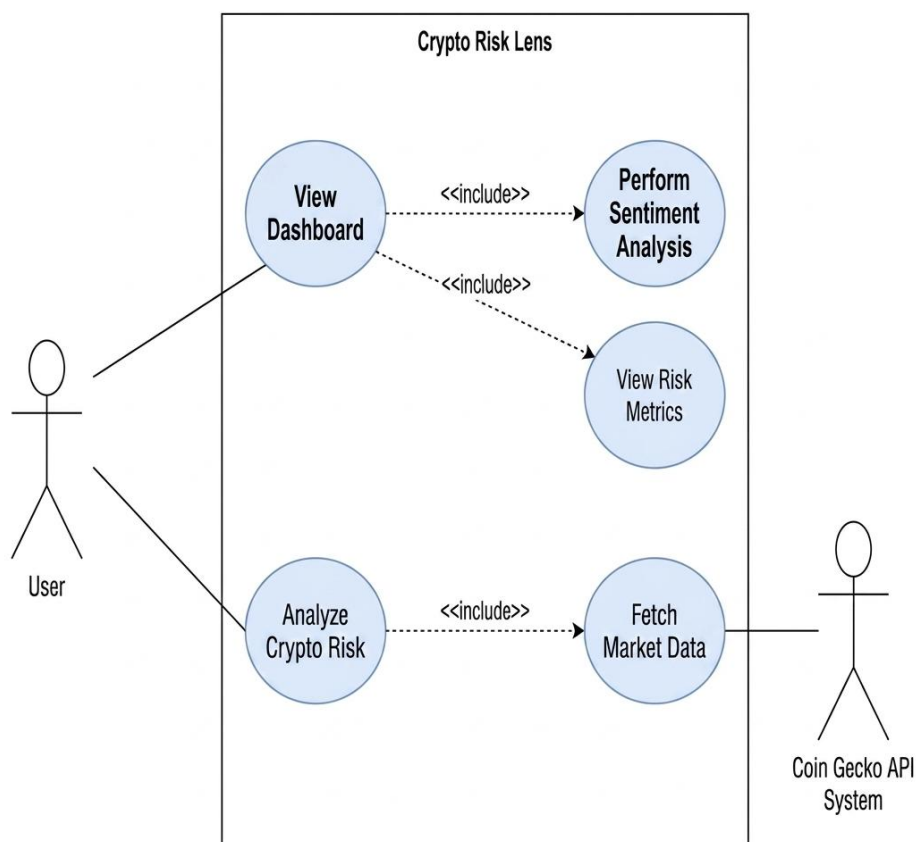


fig 5.1. Use Case Diagram

The figure 5.1 represents the use case diagram of the Crypto Risk Lens system, showing how users, internal system components, and external data sources interact to perform real-time cryptocurrency risk analysis and generate market insights. It highlights the functional relationships between different actors and explains how the system integrates data processing, machine learning, and sentiment analysis to deliver accurate and meaningful results.

The system is primarily designed for users who want to analyse cryptocurrency markets. The main actor is the user, while an external entity, the CoinGecko API, provides real-time market data such as price, volume, and trends. The system combines this data with machine learning models and sentiment analysis techniques to produce insights related to risk, market behavior, and sentiment dynamics.

At the user interaction level, the user accesses the dashboard and requests analysis for a specific cryptocurrency. The system then displays important insights such as price trends, sentiment results, and risk metrics, helping the user understand market conditions and make better decisions. The CoinGecko API plays a critical role by continuously supplying real-time data required for accurate predictions and analysis.

Internally, the system consists of multiple processes working together. The dashboard module presents all insights in a unified and easy-to-understand format. The sentiment analysis module processes textual data from sources like news articles and social media using NLP models such as CryptoBERT to classify sentiment as bullish, bearish, or neutral. The risk metrics module provides detailed outputs like risk levels, volatility, confidence scores, and probability distributions. The core analysis module applies machine learning models such as XGBoost, Hidden Markov Models, K-Means clustering, and Quantile Regression to detect patterns, forecast volatility, and classify risk. Additionally, the system includes a data fetching module that retrieves real-time data from the CoinGecko API for further processing.

Each actor in the system has a specific role. The user interacts with visualization and analysis features such as viewing dashboards, checking sentiment, and analyzing risk. The internal system components handle data processing, machine learning computations, and result generation. The external CoinGecko API acts as the main data provider, supplying the essential market data required for analysis.

Overall, the system follows a modular and scalable design that combines real-time data, machine learning and sentiment analysis for accurate crypto risk analysis.

5.2. DFD (Data Flow Diagram)

This data flow diagram (fig 5.2.1 and fig 5.2.2) for Crypto Risk Lens shows the logical, modular design for a robust cryptocurrency risk analysis system, optimizing for accurate market insights and actionable, real-time decision support for users.

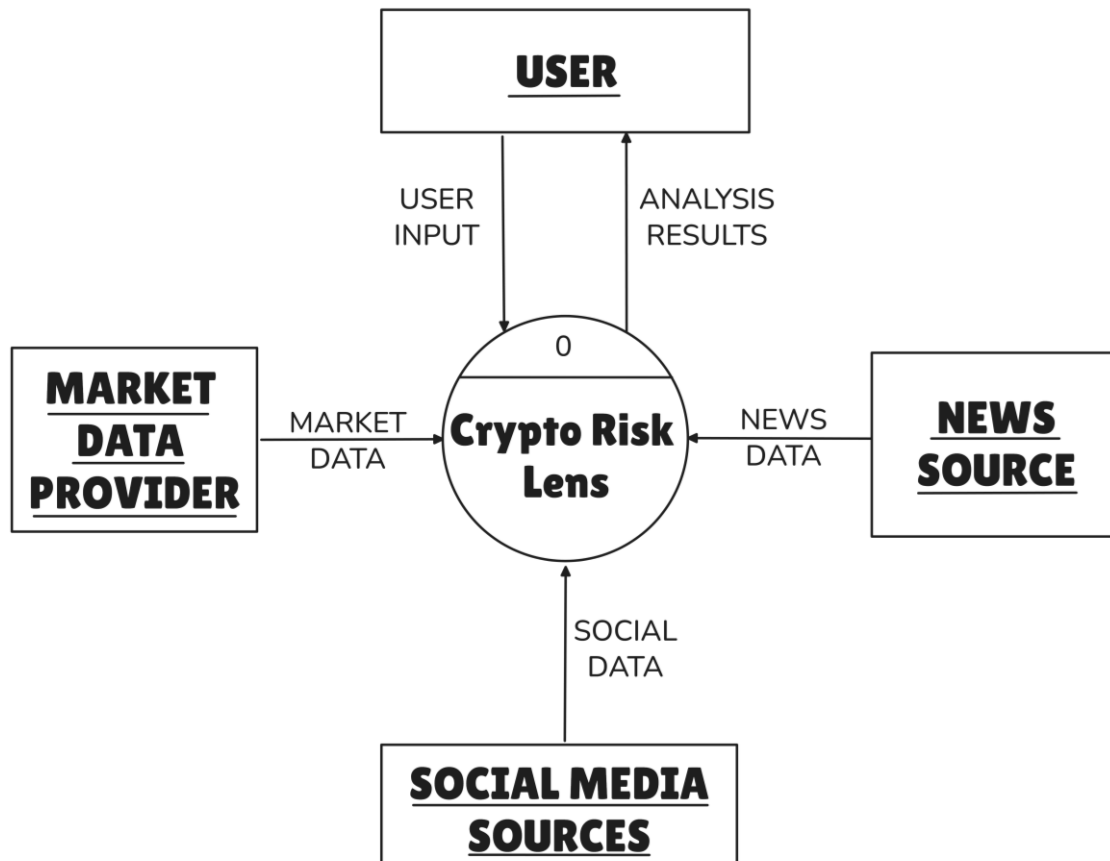


fig 5.2.1 Level 0 - Data Flow Diagram– Crypto Risk Lens

The figure 5.2.1 Level 0 DFD presents a high-level overview of the entire system, showing how data flows between external entities, the core system, and the outputs. At the center is the Crypto Risk Lens System, which acts as the main processing unit. It is responsible for collecting, integrating, and analyzing data from multiple sources such as market data, news, and social media. The system applies machine learning models and analytical techniques to generate risk predictions, sentiment insights, and overall market analysis.

The system receives input from several external entities. Market Data Providers supply real-time and historical cryptocurrency data, including price, trading volume, and market trends. News Sources provide financial articles that are used for sentiment analysis. Social

Media Sources contribute user-generated content such as opinions and discussions, which help in understanding market sentiment. Users also act as external entities by interacting with the system, entering cryptocurrency queries, and requesting risk analysis.

Once the data is received, the system processes it by integrating inputs from all sources. It performs feature extraction, applies sentiment analysis techniques, and uses machine learning models to generate meaningful insights and risk assessments.

The output of the system includes risk classification results, volatility insights, sentiment scores, and overall market analysis. These results are presented to users through an interactive dashboard and can also assist in informed decision-making.

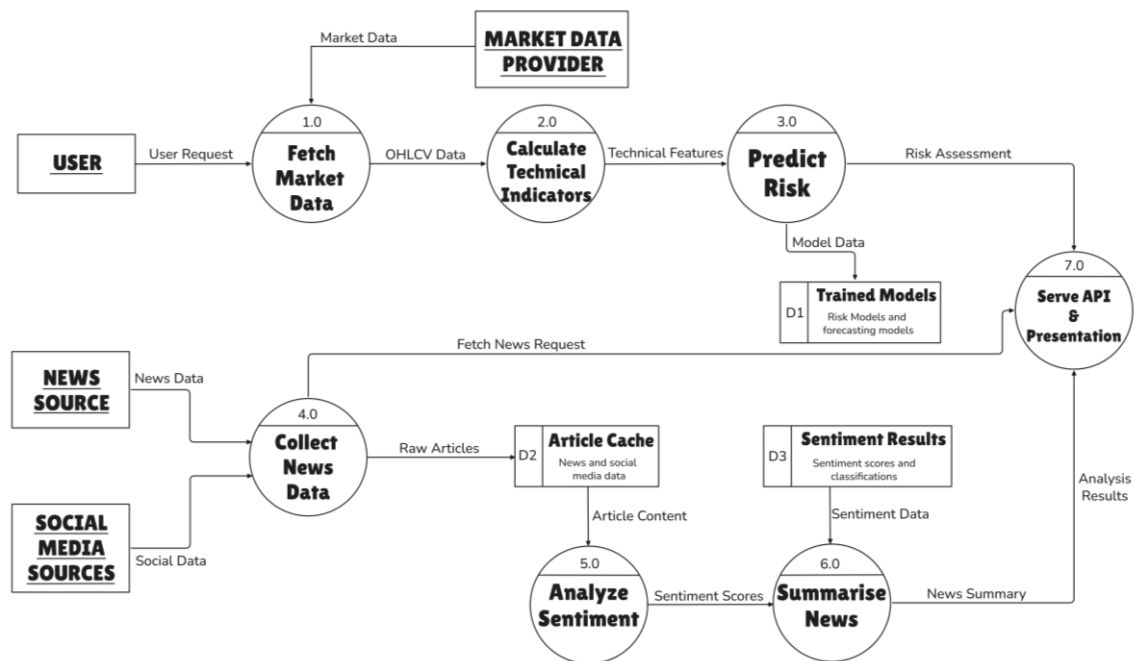


fig 5.2.2 Level 1 - Data Flow Diagram– Crypto Risk Lens

The figure 5.2.2 Level 1 DFD explains how the main system is broken into key components and how data flows step by step through each stage. The first component is the Market Data Processing Service, which fetches real-time and historical cryptocurrency data based on user queries from market data providers. It processes OHLCV (Open, High, Low, Close, Volume) data and converts it into a structured and consistent format so that it can be used for further analysis.

The next component is the Technical Indicator Engine, which takes the processed market data and calculates important technical indicators such as RSI, MACD, volatility, and

trend-based features. This step transforms raw numerical data into meaningful features that are required as input for machine learning models, especially for risk prediction.

After this, the Risk Prediction System uses trained machine learning models like XGBoost and other forecasting techniques to classify the risk level of the cryptocurrency. Based on the technical features, it generates predictions and assigns categories such as low, medium, or high risk along with additional predictive insights.

Parallel to this, the Sentiment Analysis Pipeline collects textual data from news sources and social media platforms. It processes and stores this data efficiently, then applies Natural Language Processing techniques to analyze sentiment. The system generates sentiment scores, classifies them as positive, negative, or neutral, and also summarizes the news content to provide clear and concise insights.

Finally, the API and Presentation Layer integrates the outputs from both the risk prediction system and the sentiment analysis pipeline. It serves the processed data through APIs to the frontend dashboard, where results such as risk levels, sentiment scores, and summaries are displayed in an interactive and user-friendly format.

Additionally, the system includes a feedback mechanism where trained models are continuously updated and reused, allowing the system to improve its performance over time. This structured flow ensures efficient real-time data processing, accurate risk assessment, and effective visualization of cryptocurrency market insights.

The diagram also highlights clear dependencies between modules, where risk prediction relies on computed technical indicators and market data, while the dashboard integrates both sentiment and risk outputs for complete analysis. Data storage components like article cache, sentiment results, and trained models ensure efficient reuse of processed data and improve system performance. This structured flow enables seamless coordination between data collection, processing, prediction, and presentation for reliable real-time insights. Each stage of the pipeline is loosely coupled, making the system easy to scale, maintain, and upgrade without affecting other components. This design improves overall reliability and ensures that the system can handle large volumes of data efficiently.

5.3. System Architecture

This architecture diagram (fig 5.3.) provides a systematic, modular overview of how the "Crypto Risk Lens" system works from input to output, highlighting both data and model flow. Here we breaks down the system into clear components: Input, Frontend Layer, Backend Processing, Modeling, Sentiment Pipeline, and Output.

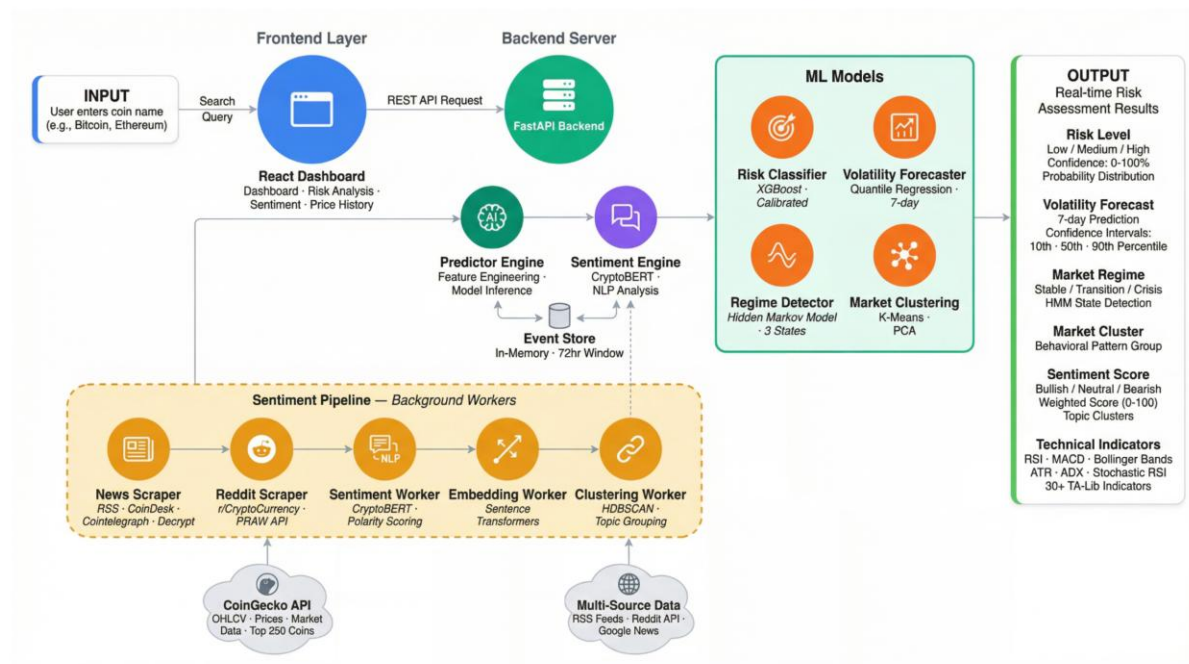


fig 5.3: System Architecture

The system begins with the input layer, where users enter a cryptocurrency name such as Bitcoin or Ethereum through the interface to initiate analysis. Along with this, real-time market data including price, volume, and OHLCV is fetched using APIs like CoinGecko. Additional external data sources such as news articles, Reddit posts, RSS feeds, and sentiment indicators like the Fear & Greed Index are also collected to enrich the analysis.

In the frontend layer, a React-based dashboard provides an interactive interface where users can view risk analysis, sentiment insights, and price history. It also manages user interactions by handling search queries and sending REST API requests to the backend for further processing.

The backend processing is handled by a FastAPI server, which manages API requests and coordinates data flow between different components. Within this layer, the predictor engine performs feature engineering and model inference, while the sentiment engine applies

NLP techniques such as CryptoBERT to analyze textual data and extract sentiment information. An event store is used to temporarily hold processed data, enabling efficient real-time updates and computations.

The modeling layer consists of multiple machine learning models designed for different analytical tasks. A risk classifier based on XGBoost categorizes cryptocurrencies into low, medium, or high risk. A volatility forecaster using quantile regression predicts future price volatility along with confidence intervals. A regime detector based on Hidden Markov Models identifies market conditions such as stable, transition, or crisis states. Additionally, market clustering is performed using K-Means combined with PCA to group similar market behaviors and detect patterns.

The sentiment pipeline operates through background workers that continuously process textual data. News and Reddit scrapers collect real-time information, which is then analyzed by a sentiment worker to determine sentiment polarity. An embedding worker converts text into numerical representations using transformer models, and a clustering worker applies DBSCAN to group similar topics and discussions for deeper insights.

Finally, the output layer presents the results in a structured format. It includes a risk assessment showing the risk level with confidence scores and probability distribution, a volatility forecast with percentile-based intervals, and identification of the current market regime. It also displays market clusters, sentiment scores indicating bullish, neutral, or bearish trends, and technical indicators such as RSI, MACD, Bollinger Bands, and ATR for comprehensive analysis.

The architecture also emphasizes real-time processing by integrating streaming data with background workers, ensuring continuous updates without blocking the main system flow.

Loose coupling between components allows independent scaling of modules like sentiment analysis and model inference, improving performance under high data loads. This design ensures robustness, faster response times, and the ability to extend the system with additional models or data sources in the future. It also supports efficient fault isolation, where failures in one module do not disrupt the entire system. This improves system stability and ensures consistent delivery of accurate insights even under heavy usage.

5.4. Implementation

The Implementation diagrams presented for the Crypto Risk Lens system visualize different parts of the deployed user interface and data-driven functionality of the cryptocurrency risk analysis platform. Each diagram represents a specific component of the system, illustrating how real-time market data, sentiment analysis, and machine learning models are integrated to provide meaningful insights into cryptocurrency risk and market behavior.

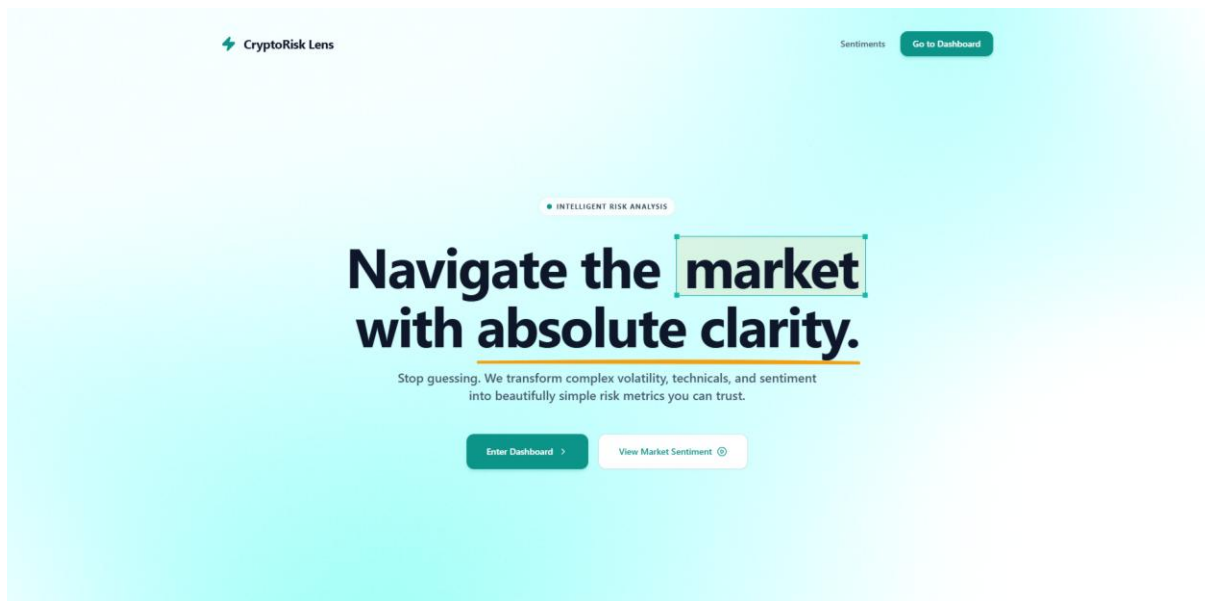


fig 5.4.1: Crypto Risk Lens Landing Page

The figure 5.4.1 depicts the landing page which introduces the Crypto Risk Lens system, which helps users understand market risk in a simple and clear way. The main idea is to remove confusion from crypto trading by combining market data, technical indicators, and sentiment analysis into easy-to-read risk insights. The page highlights that users do not need to guess market trends, as the system processes complex information and presents it in a clear format. It also provides quick actions like entering the dashboard to view detailed analytics or checking market sentiment. Overall, the landing page communicates that the platform turns complicated market behavior into simple, reliable risk metrics for better decision-making.

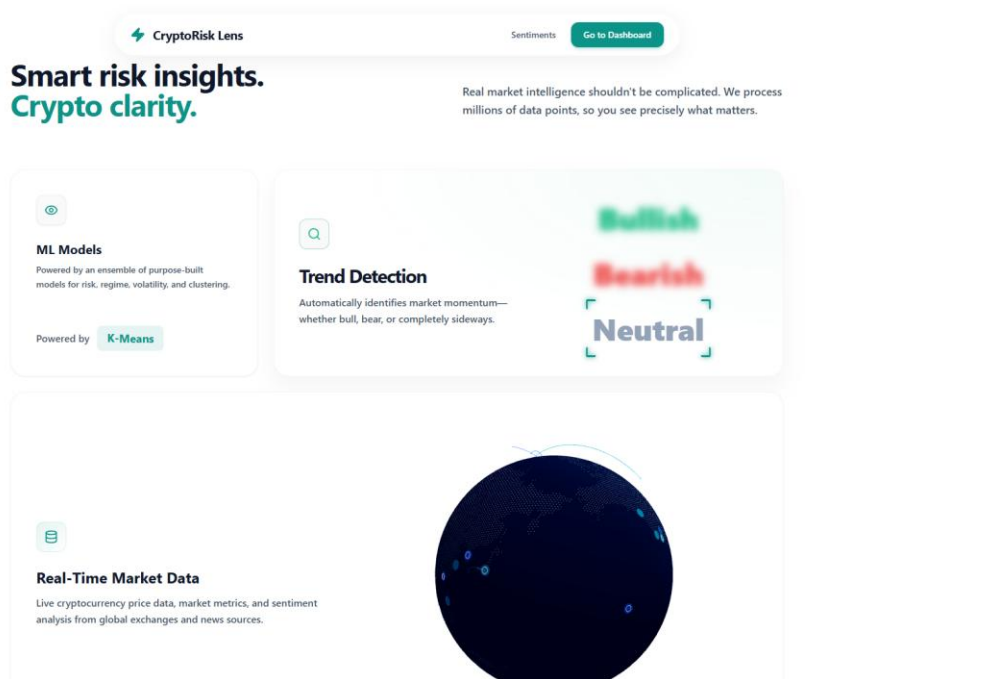


fig 5.4.2: The Crypto Risk Lens Core Architecture & Intelligence Framework

The figure 5.4.2 the Crypto Risk Lens Core Architecture & Intelligence Framework explains how the system generates clear and meaningful market insights. It utilizes machine learning models such as Quantile Regression, Hidden Markov Models (HMM), and XGBoost to analyze risk, volatility, and market patterns. The system automatically detects trends and classifies the market as bullish, bearish, or neutral. It also collects real-time data from cryptocurrency prices, global exchanges, and news sources to ensure up-to-date analysis. By combining these components, the system processes large volumes of data and converts them into simple, accurate insights for understanding market conditions.

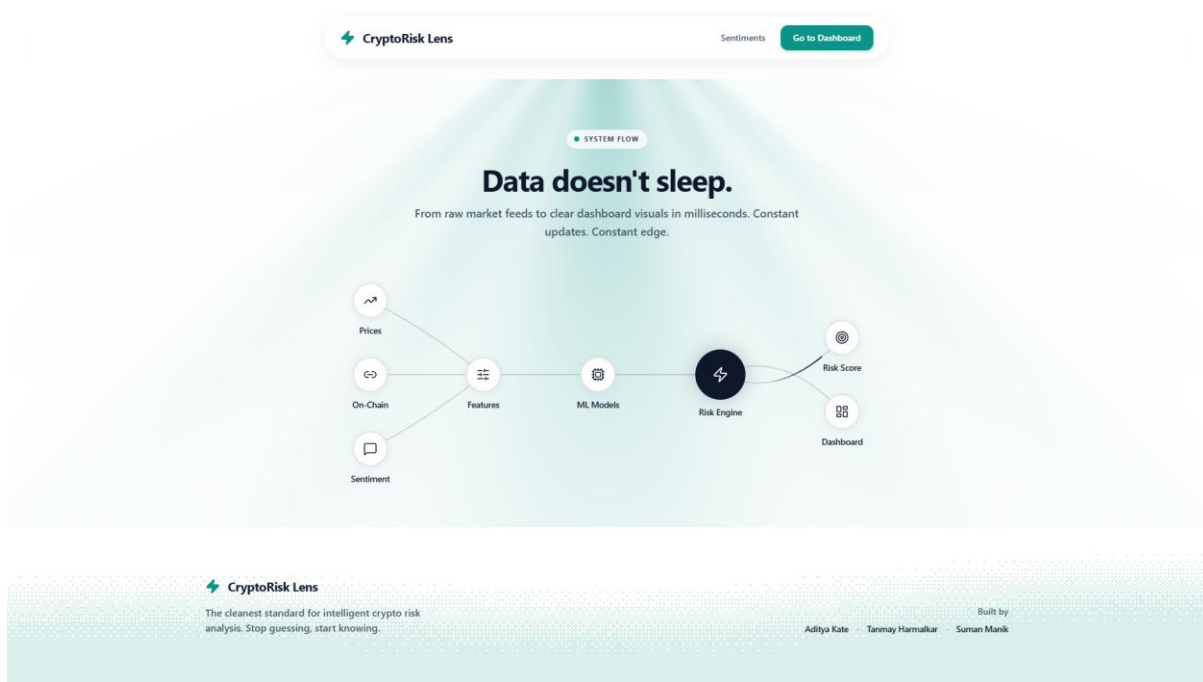


fig 5.4.3: Crypto Risk Lens System Workflow and Data Processing Pipeline

The figure 5.4.3 Crypto Risk Lens System Workflow and Data Processing Pipeline illustrates the working flow of the system. It begins with the collection of multiple data sources, including price data, on-chain data, and market sentiment. These inputs are processed and transformed into meaningful features, which are then passed to machine learning models for analysis. The analyzed data is handled by a risk engine that computes the final risk score. The results are subsequently displayed on the dashboard in a clear and visual format for easy interpretation. The entire system operates continuously, ensuring fast updates and providing real-time insights to users.

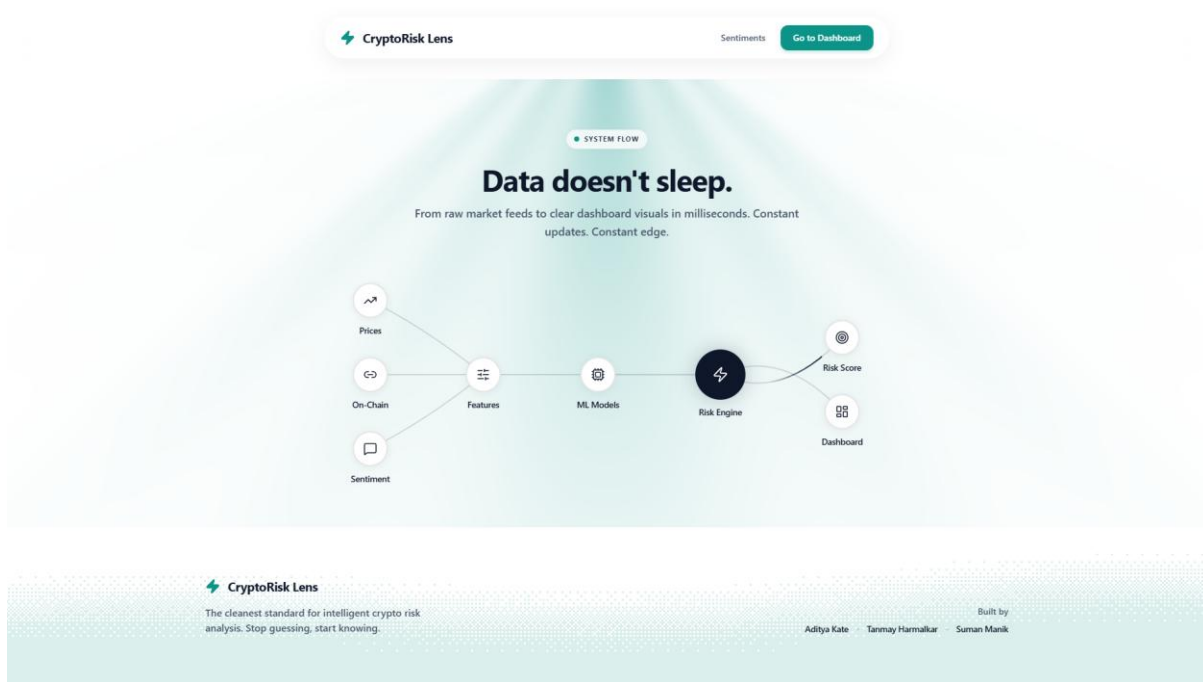


fig 5.4.4: Crypto Risk Lens Dashboard

The figure 5.4.4 shows the Crypto Risk Lens dashboard which provides a real-time overview of the cryptocurrency market, focused on Bitcoin (or any other coin that can be searched), displaying key metrics such as current price, 24-hour trading volume, market capitalization, and short-term volatility, along with a price chart that shows recent market trends over time to help users understand movement patterns. The risk analysis section calculates the current risk score and classifies it (e.g., speculative), along with model confidence, and also presents risk distribution by categorizing conditions into low, medium, and high risk. Overall, the dashboard combines live data and model outputs to deliver clear, data-driven insights for monitoring market conditions and risk levels.



fig 5.4.5: 7-Day Cryptocurrency Price Movement (Candlestick Chart)

The figure 5.4.5 illustrates the 7-day price movement of Bitcoin using a candlestick representation, providing a detailed view of short-term market behaviour. Each candlestick represents price fluctuations within a specific time interval, where green candles indicate a price increase and red candles indicate a decrease. The chart allows users to easily observe market trends, such as upward or downward movements, and identify important patterns like consolidation, breakouts, and reversals over time. By visually representing both price direction and volatility, the figure helps users gain deeper insights into market dynamics and trading activity. Overall, it serves as an effective tool for analyzing short-term performance, understanding market momentum, and supporting data-driven decision-making in cryptocurrency analysis.

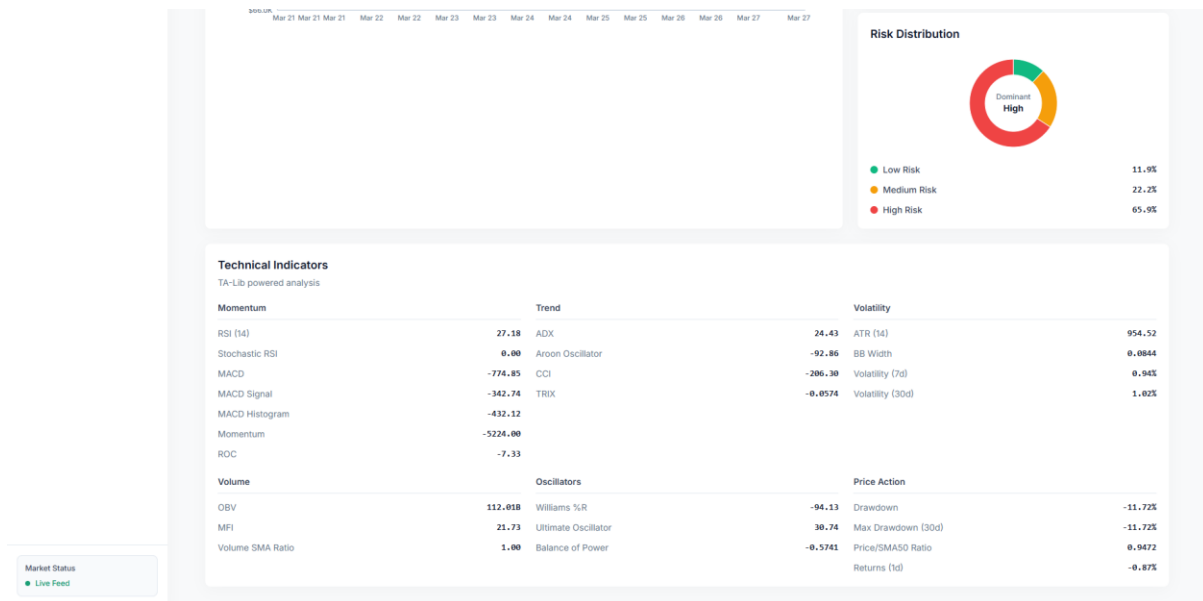


fig 5.4.6: Crypto Risk Lens Comprehensive Technical Indicator Matrix

The figure 5.4.6 presents a detailed technical analysis of the cryptocurrency market using multiple indicators to evaluate price behavior and risk. It includes momentum indicators such as RSI, MACD, and ROC to measure the strength and direction of price movement, while trend indicators like ADX and Aroon help determine whether the market is trending or moving sideways. Volatility indicators, including ATR and Bollinger Band width, illustrate the extent of price fluctuations over time. In addition, volume indicators capture trading activity, and oscillators assist in identifying overbought or oversold conditions. The analysis also incorporates price action metrics such as drawdown and returns to assess recent performance. Together, these indicators provide a comprehensive understanding of market dynamics and support more accurate and data-driven risk evaluation.

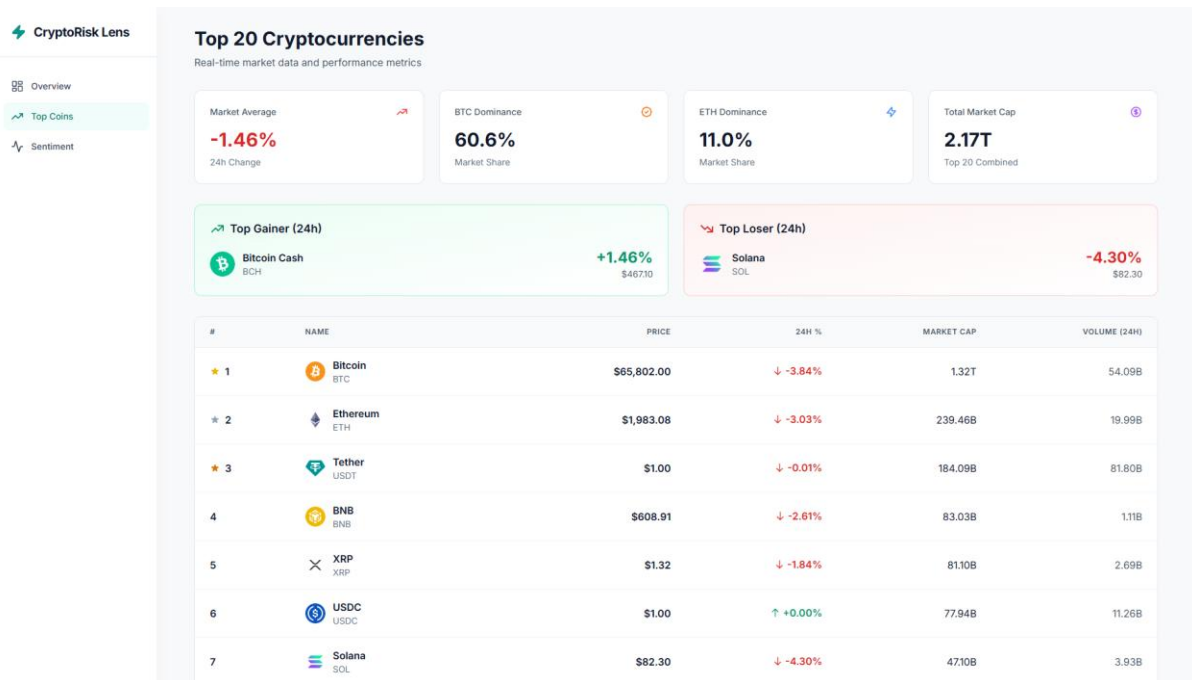


fig 5.4.7: Crypto Risk Lens Global Market Overview and Top Assets Leaderboard

The figure 5.4.7 presents a real-time overview of the top 20 cryptocurrencies based on their market performance, providing a comprehensive snapshot of current market conditions. It includes key summary metrics such as overall market change, Bitcoin and Ethereum dominance, and total market capitalization, offering insights into the broader market structure. The figure also highlights the top gaining and top losing cryptocurrencies over the last 24 hours, enabling users to quickly identify significant market movements. Additionally, a detailed table displays each cryptocurrency along with its current price, 24-hour percentage change, market capitalization, and trading volume. This structured representation allows users to compare different assets efficiently and gain a clear understanding of overall market trends.

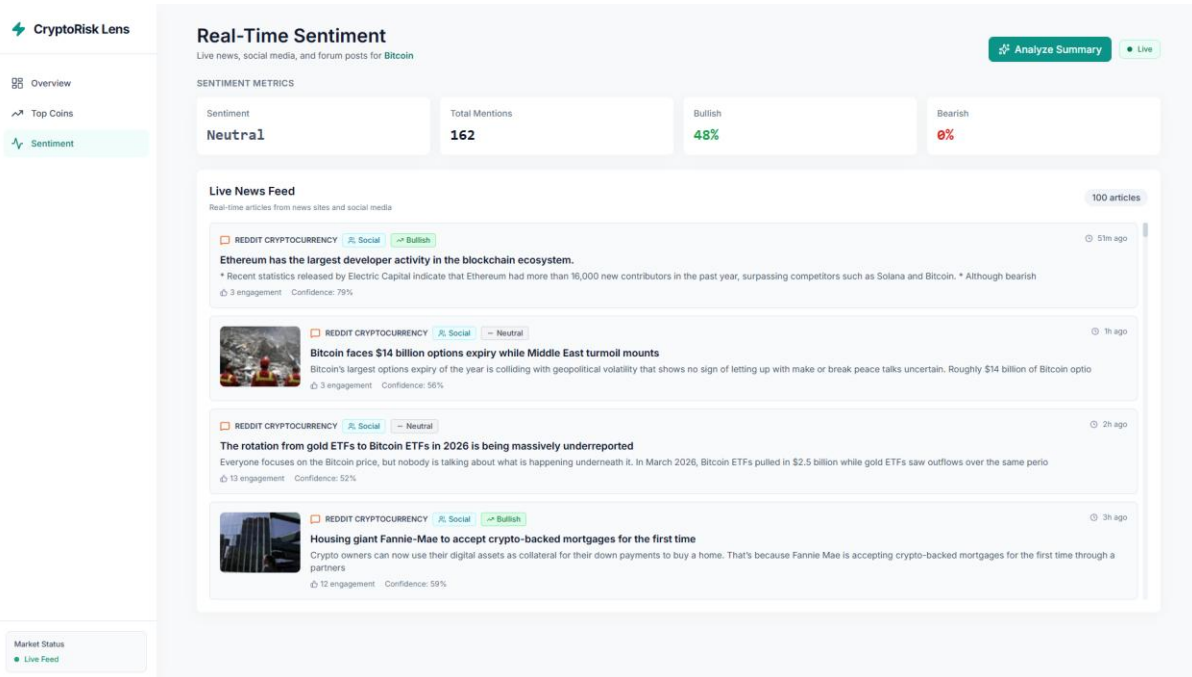


fig 5.4.8: Crypto Risk Lens Real-Time Sentiment Analysis and News Feed

The figure 5.4.8 presents real-time sentiment analysis data collected from multiple sources such as RSS feeds, news websites, blogs, and social media platforms like Reddit, offering a comprehensive view of market perception. It displays key metrics including overall market sentiment, total mentions, and the proportion of bullish and bearish opinions, providing a quick summary of public sentiment. A live feed section lists recent articles and posts along with their sentiment labels, engagement levels, and confidence scores for deeper analysis. Additionally, when the user selects the “Analyze Summary” option, the system processes the collected data and generates a concise summarized sentiment report for the chosen cryptocurrency, enabling users to better understand the overall market outlook and sentiment trends.

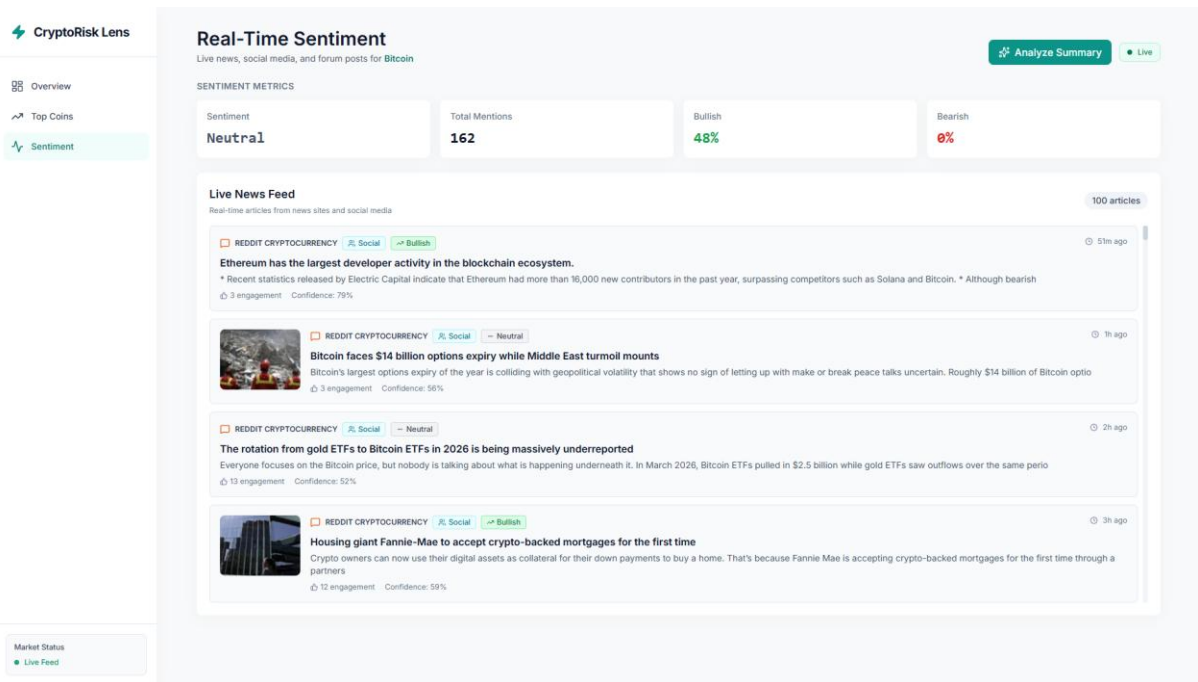


fig 5.4.9: Crypto Risk Lens Automated News Summary and NLP Sentiment Detail

The figure 5.4.9 illustrates the detailed view of a selected article or post within the sentiment analysis module. When a user clicks on any article, a comprehensive view is displayed showing the full content along with important metadata such as the source (e.g., Reddit or a news website), sentiment label, timestamp, and engagement details. The content is presented in a clear and structured format, often highlighting key points to improve readability and understanding. Additionally, a direct source link is provided, allowing users to access the original article for further reference. This feature enables users to analyze the information in depth and better understand the context behind the sentiment analysis.

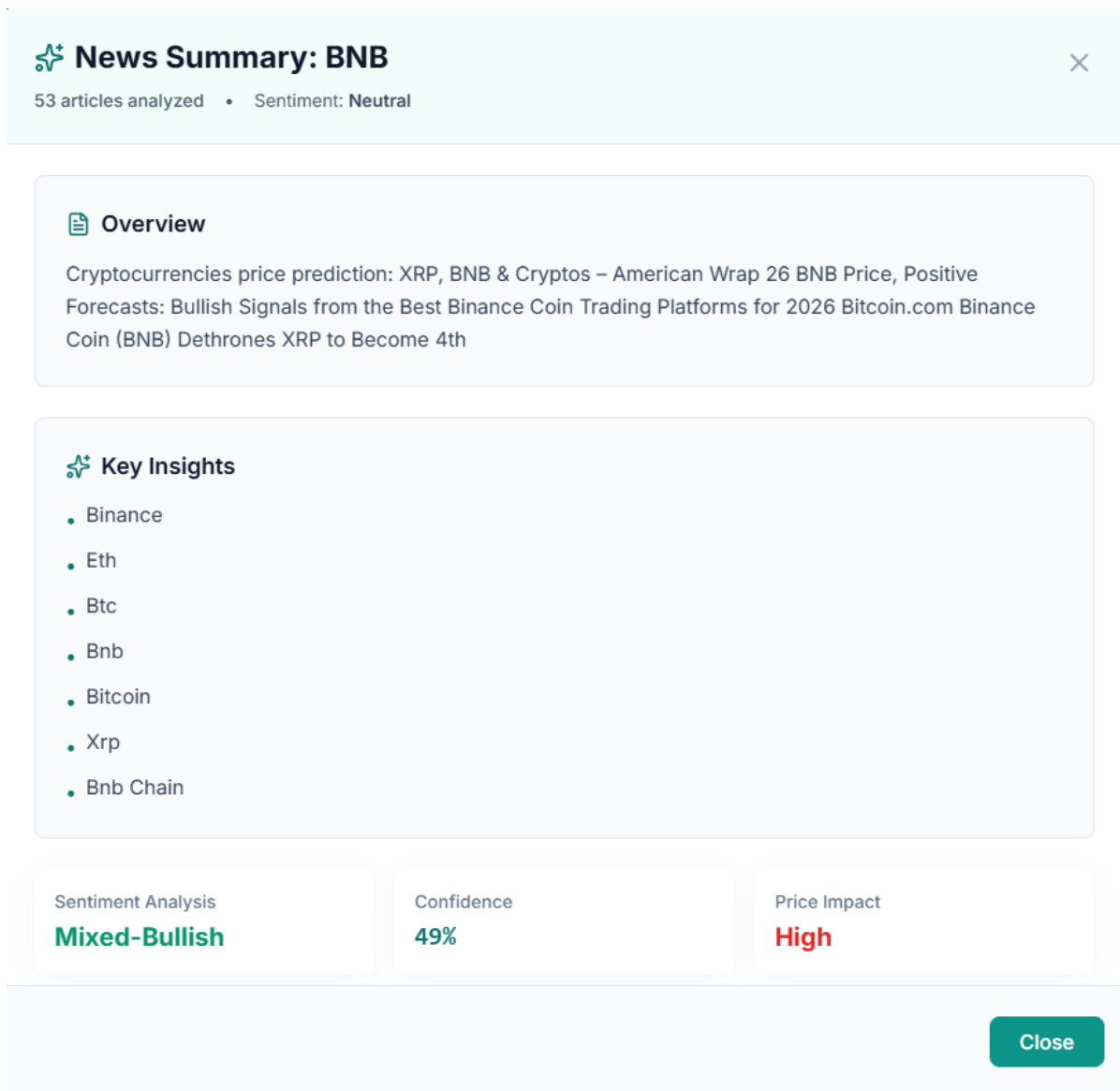


fig 5.4.10: Crypto Risk Lens Aggregate News Sentiment and Market Analysis

The figure 5.4.10 illustrates the Aggregate News Summary interface, which synthesizes data from a large volume of articles in this instance, 53 articles analyzed for BNB. The module provides a high-level Overview of recent headlines and extracts Key Insights by identifying major entities and themes relevant to the asset. At the base of the interface, the system outputs a nuanced Sentiment Analysis (e.g., Mixed-Bullish) accompanied by a Confidence percentage. Unique to this view is the Price Impact indicator, which evaluates the potential severity of these news events on the asset's market value. This component is essential for filtering through massive datasets to identify high-impact narratives that could drive short-term volatility.

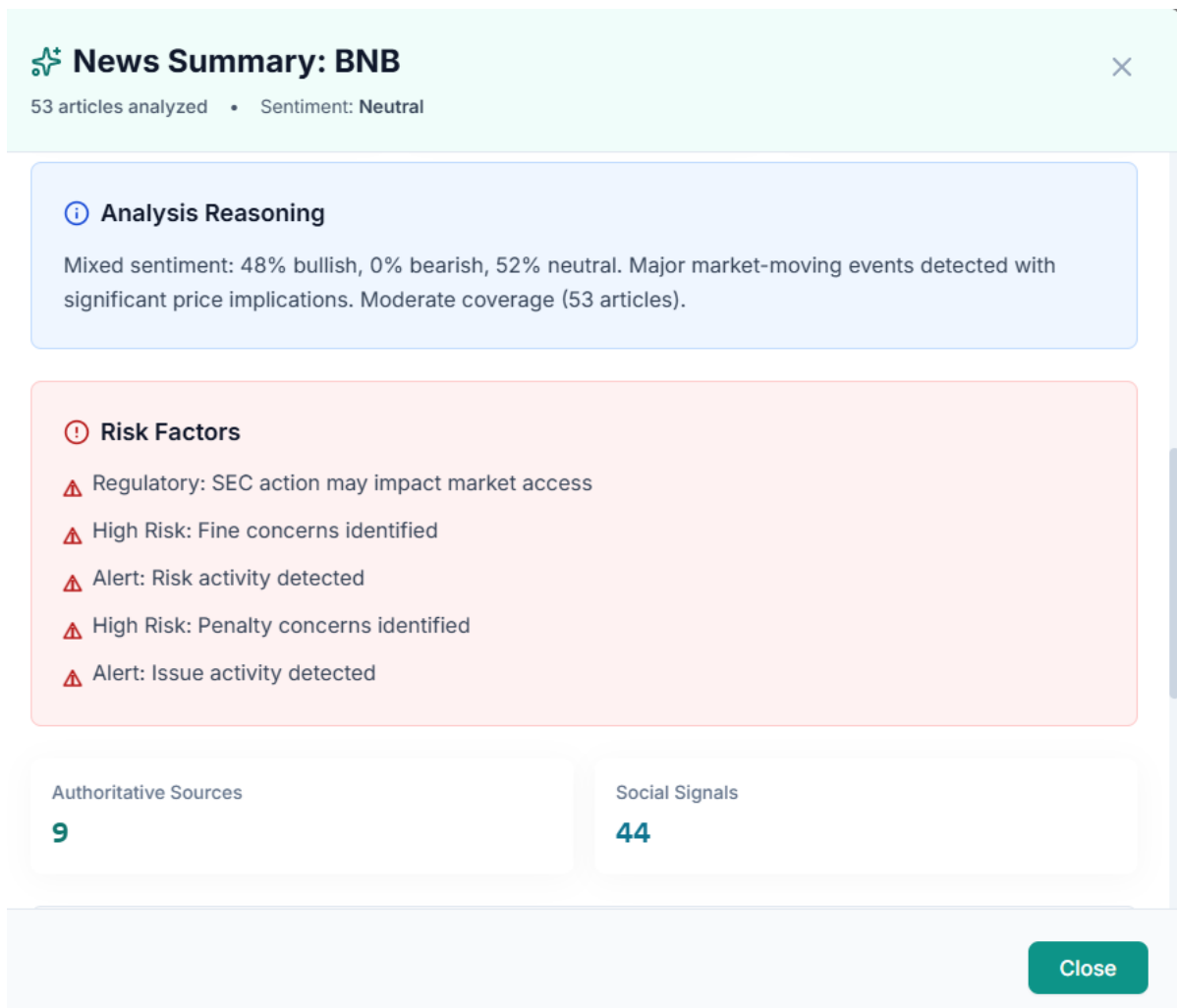


fig 5.4.11: Crypto Risk Lens News Summary Analysis and Risk Factor Assessment

The figure 5.4.11 illustrates the Analysis Reasoning and Risk Factor diagnostic interface for a specific asset in this case, BNB based on a sample of 53 articles. The system provides a transparent breakdown of its sentiment logic, showing the percentage distribution between bullish, bearish, and neutral tones to justify the final classification. Below this, a high-priority Risk Factors section flags critical qualitative threats, such as potential regulatory actions (e.g., SEC involvement), financial penalties, or unusual risk activities detected within the news cycle. The interface also quantifies the source reliability by displaying the number of Authoritative Sources versus Social Signals. This multi-layered approach ensures that the risk score is grounded in both statistical sentiment and specific, high-impact external events.

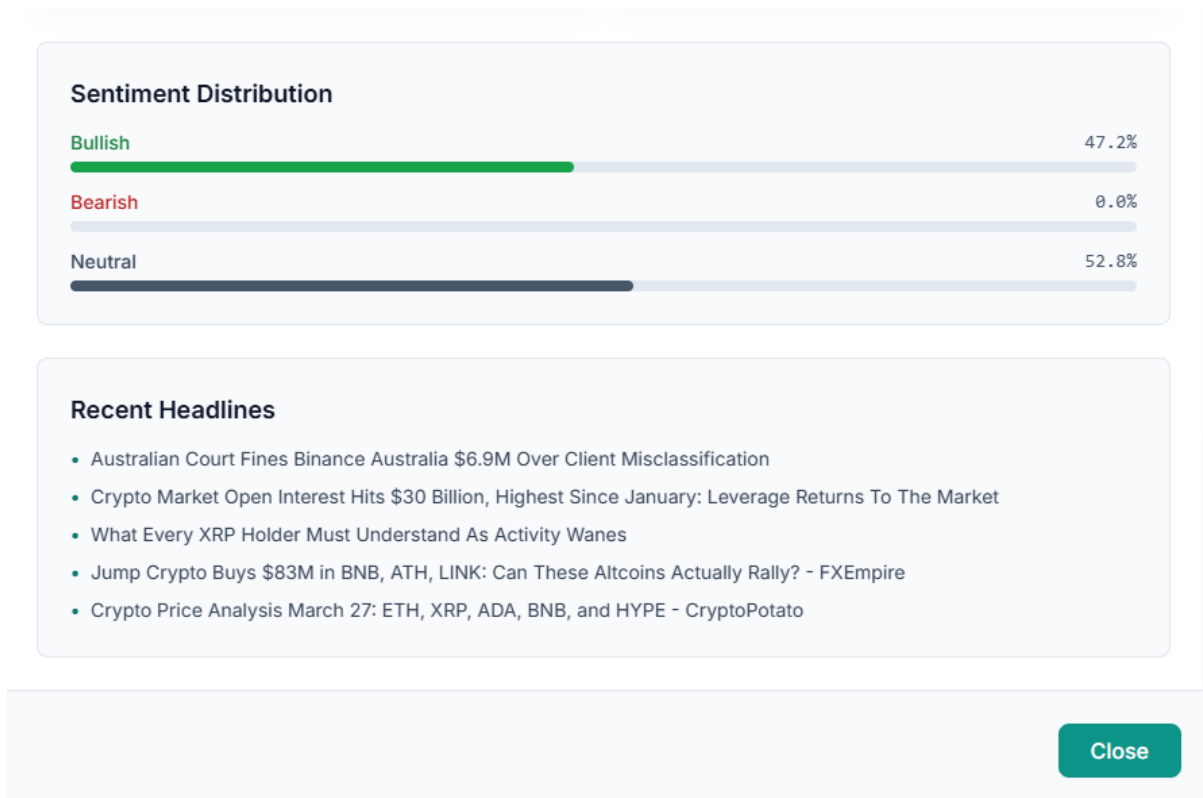


fig 5.4.12: Crypto Risk Lens Sentiment Distribution and Recent Market Headlines

The figure 5.4.12 presents the Sentiment Distribution breakdown and a list of Recent Headlines impacting the risk profile of the analyzed asset. The sentiment distribution provides a quantitative visualization of market mood, displaying precise percentages for Bullish (47.2%), Bearish (0.0%), and Neutral (52.8%) categories. Below this, the Recent Headlines section lists current, high-impact news stories-ranging from regulatory fines and institutional buys to general price analysis-collected from diverse financial news sources. This module allows the system to ground its risk assessments in verifiable events while giving the user direct access to the specific narratives driving the current market sentiment and risk score.

Chapter 6

Technical Specification

The technical specifications of the Crypto Risk Lens project outline the core architectural components, tools, and technologies used for its development and operation. This section provides a comprehensive overview of the system architecture, data processing techniques, machine learning models, and software technologies integrated into the system. It includes details about the frameworks, APIs, data sources, and visualization tools used, along with considerations for performance, scalability, and usability.

Technical specifications are essential for understanding how the system is designed, implemented, and maintained. They provide insights into the system's efficiency, reliability, and ability to handle real-time data processing. This documentation serves as a reference for developers and stakeholders, ensuring that the system meets its functional requirements and can be extended or improved in the future.

Frontend:

1. React with TypeScript
2. Recharts.js
3. D3.js

Technologies: The frontend of Crypto Risk Lens is developed using React and TypeScript, enabling the creation of a dynamic, responsive, and interactive user interface for real-time cryptocurrency analysis.

Visualization Libraries: Recharts and D3.js are used to display charts and graphs for price trends, volatility, sentiment scores, and risk levels, providing clear and interactive data visualization.

Backend Development:

1. Python 3.10
2. FastAPI/Flask

Framework: Python serves as the primary backend language, with FastAPI handling API development, data processing, and communication between the frontend and backend. It supports asynchronous operations for efficient real-time data handling.

Libraries and Tools:

1. Data Handling and Processing: NumPy and Pandas are used for data preprocessing and management, ensuring clean integration of historical and real-time datasets.

2. Machine Learning: Scikit-learn is used to implement models such as Random Forest and K-Means clustering, while XGBoost is applied for advanced gradient boosting-based risk prediction. Libraries such as hmmlearn or statsmodels are used for Hidden Markov Models to detect market regimes.

3. Visualization and Analysis: Recharts and D3.js are used for interactive frontend visualizations, while Matplotlib is utilized for backend analysis and plotting. APIs such as CoinGecko provide real-time cryptocurrency data, and RSS feeds along with social media sources supply sentiment data.

Algorithms:

1. Random Forest Classifier:

The Random Forest Classifier serves as a core machine learning model for predicting cryptocurrency risk levels. This ensemble learning technique constructs multiple decision trees during training and outputs the majority class for classification.

The algorithm analyzes historical and real-time data such as price, trading volume, volatility, and technical indicators. Each tree evaluates different features, enabling the model to classify assets into risk categories such as Low, Medium, and High Risk. By aggregating predictions from multiple trees, Random Forest reduces overfitting and improves overall prediction accuracy while providing confidence scores for each prediction. Additionally, the model provides confidence scores for each prediction, offering insights into the certainty of risk classification. Its robustness against noisy data and ability to handle large feature sets make it well-suited for real-time cryptocurrency risk analysis, where market conditions are highly dynamic and unpredictable.

2. XGBoost:

XGBoost is used as an advanced machine learning model for improving prediction accuracy and handling complex market patterns. It builds models sequentially, where each new model corrects errors made by previous ones.

The algorithm processes multiple financial indicators and market features, optimizing performance through gradient boosting techniques. It is highly efficient, handles large datasets effectively, and provides better generalization compared to traditional models, making it suitable for real-time risk prediction. Additionally, XGBoost is designed for high performance and efficiency, supporting parallel processing and handling large-scale datasets with ease. Its ability to generalize well across unseen data, combined with fast computation speed, makes it particularly suitable for real-time risk prediction and analysis in fast-changing financial environments.

3. K-means Clustering:

K-Means Clustering is used to group cryptocurrencies or market conditions into distinct clusters based on similar characteristics. This unsupervised learning technique helps identify hidden patterns in market behavior without requiring labeled data.

The algorithm analyzes features such as price trends, volatility, trading volume, and technical indicators. It partitions data into clusters where each group represents similar market behavior, enabling better understanding of trends such as bullish, bearish, or stable conditions. This clustering approach enables the system to identify different market conditions such as bullish trends, bearish phases, or stable periods. By grouping similar patterns, it provides valuable insights into market segmentation and helps improve decision-making by highlighting recurring behaviors and anomalies in the cryptocurrency market.

4. Hidden Markov Model (HMM):

The Hidden Markov Model is used for detecting underlying market regimes such as stable, transitional, and high-volatility phases. It models the market as a sequence of hidden states that evolve over time.

The algorithm uses observable data such as price returns and volatility to infer hidden states, helping in understanding transitions between different market conditions. This probabilistic approach allows the system to capture temporal dependencies and dynamic

changes in the cryptocurrency market. HMM is particularly effective for time-series analysis, as it can identify shifts in market conditions and detect patterns that are not immediately visible. This helps the system provide deeper insights into market dynamics, improve risk assessment, and support better forecasting in highly volatile cryptocurrency environments.

5. Quantile Regression:

Quantile Regression is used for volatility forecasting by estimating different quantiles of the future price distribution, providing a more comprehensive understanding of potential market movements. Unlike traditional regression methods that predict only the mean value, Quantile Regression captures the entire distribution of possible outcomes, making it highly effective for modeling financial uncertainty.

The algorithm analyzes historical price data, volatility patterns, and technical indicators to estimate lower, median, and upper quantiles of future volatility. This allows the system to identify best-case, average, and worst-case scenarios, offering a range of predictions rather than a single fixed value. Such an approach is particularly useful in highly volatile cryptocurrency markets, where extreme price movements are common. By providing prediction intervals and capturing asymmetric risk, Quantile Regression helps in understanding the spread and uncertainty of market behavior. This enables users to make more informed and risk-aware decisions, improving portfolio management and reducing exposure to unexpected market fluctuations.

Chapter 7

Project Scheduling

In project management, a schedule is a listing of a project's milestones, activities, and deliverables. The project schedule (Table 7.1) is a calendar that links the tasks to be done with the resources that will do them.

Sr.No.	Group Members	Duration	Task Performed
1.	Aditya Kate Tanmay Harmalkar Suman Manik	1 st Week of January	Group formation and topic finalization. Identifying the scope and objectives of the Crypto Risk Lens project. Discussing the project idea based on cryptocurrency risk analysis and finalizing core functionalities.
2.	Suman Manik	2 nd Week of January	Conducting literature review and studying existing systems related to cryptocurrency prediction and sentiment analysis. Exploring machine learning models such as Random Forest, XGBoost, and HMM.
3.	Suman Manik	1 st Week of February	Designing the system architecture and planning the user interface. Creating initial UI/UX designs for dashboard, charts, and sentiment analysis modules using React.
4.	Aditya Kate Tanmay Harmalkar	2 nd Week of February	Implementing data collection and preprocessing modules. Integrating APIs (CoinGecko) for real-time cryptocurrency data and preparing datasets for model training.
5.	Aditya Kate Suman Manik	1 st Week of March	Implementing machine learning models including Random Forest, XGBoost, K-Means clustering, HMM, and Quantile Regression for risk prediction, volatility forecasting, and market analysis.
6.	Aditya Kate Tanmay Harmalkar	2 nd Week of March	Testing system reliability, visualization dashboards, and overall project modules.

Table 7.1: Project Task Distribution

A Gantt chart is a type of bar chart that illustrates a project schedule. This chart lists the tasks to be performed on the vertical axis, and time intervals on the horizontal axis. Gantt chart (table 7.2) illustrates the start and finish dates of the terminal elements and summary elements of a project.

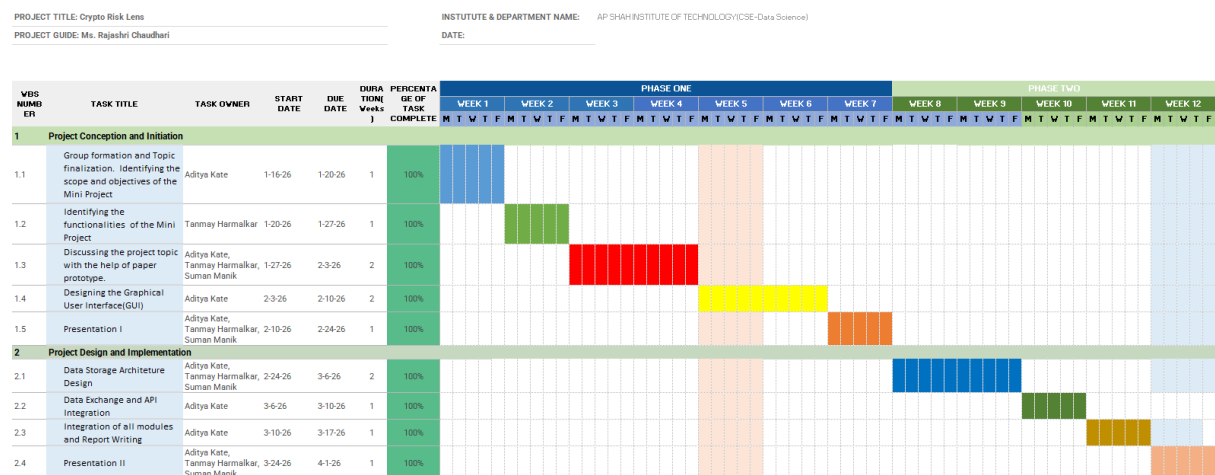


Table 7.2: Gantt Chart of Crypto Risk Lens

Chapter 8

Results

The project results section provides a concise overview of the outcomes achieved through the implementation of the project. Highlighting key findings, deliverables, and the final implementation of the project lifecycle. This section serves to summarize the tangible outcomes and impacts of the project, providing stakeholders with valuable insights into its overall effectiveness and contribution to the intended objectives.

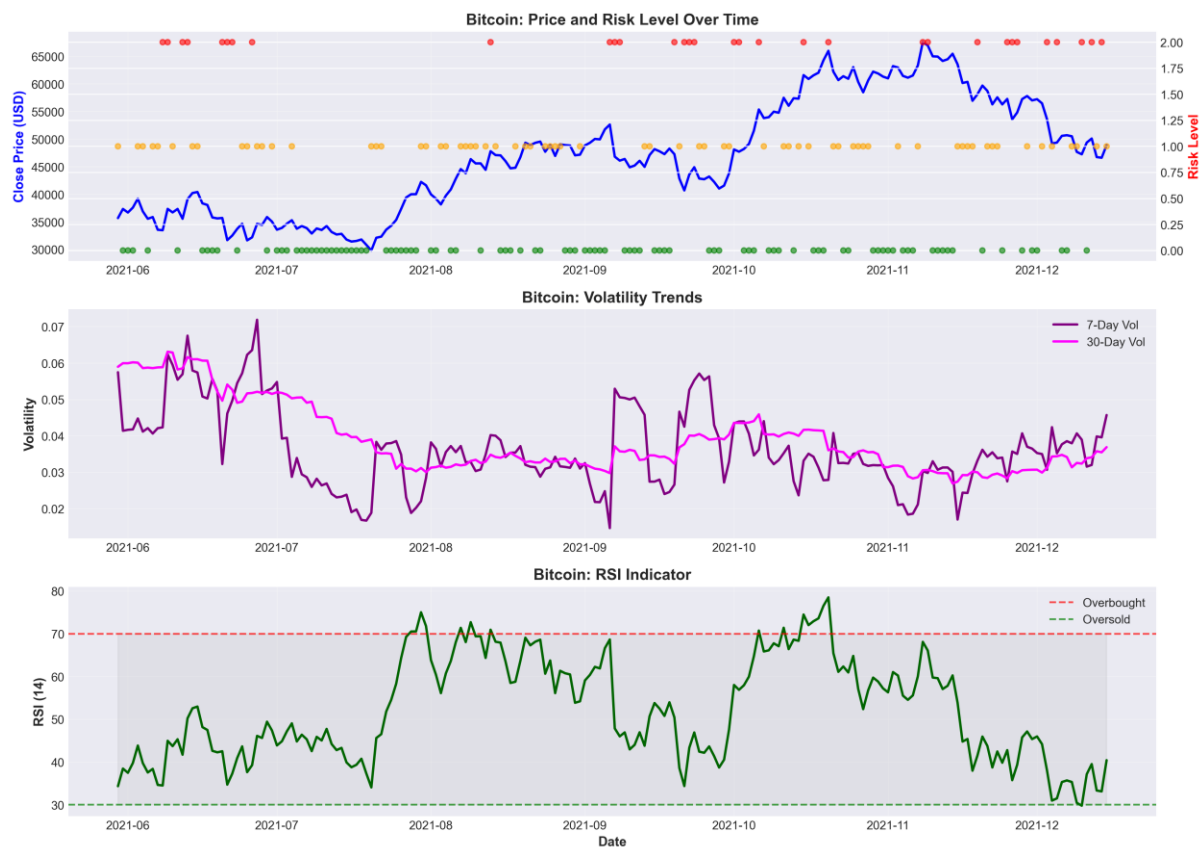


fig 8.1: Time Series Patterns Bitcoin

The figure 8.1 highlights Bitcoin's market behavior over time by combining price trends, risk levels, volatility, and technical indicators. The price generally increases from June to November before declining towards December. Risk levels vary throughout this period, with low risk shown in green, medium in orange, and high risk in red. High-risk points are mostly observed near price peaks, indicating that rapid price increases are often followed by potential corrections or instability.

The volatility analysis shows both short-term (7-day) and long-term (30-day) trends. Short-term volatility fluctuates sharply with sudden spikes and drops, while long-term

volatility remains relatively stable, indicating that Bitcoin experiences frequent short-term uncertainty despite smoother overall trends. The RSI (Relative Strength Index) reflects market momentum, typically ranging between 30 and 70. Higher RSI values near 70 indicate overbought conditions and possible price reversals, while lower values near 30 indicate oversold conditions and potential buying opportunities. Overall, these observations demonstrate the importance of combining price, volatility, and technical indicators for accurate cryptocurrency risk analysis.

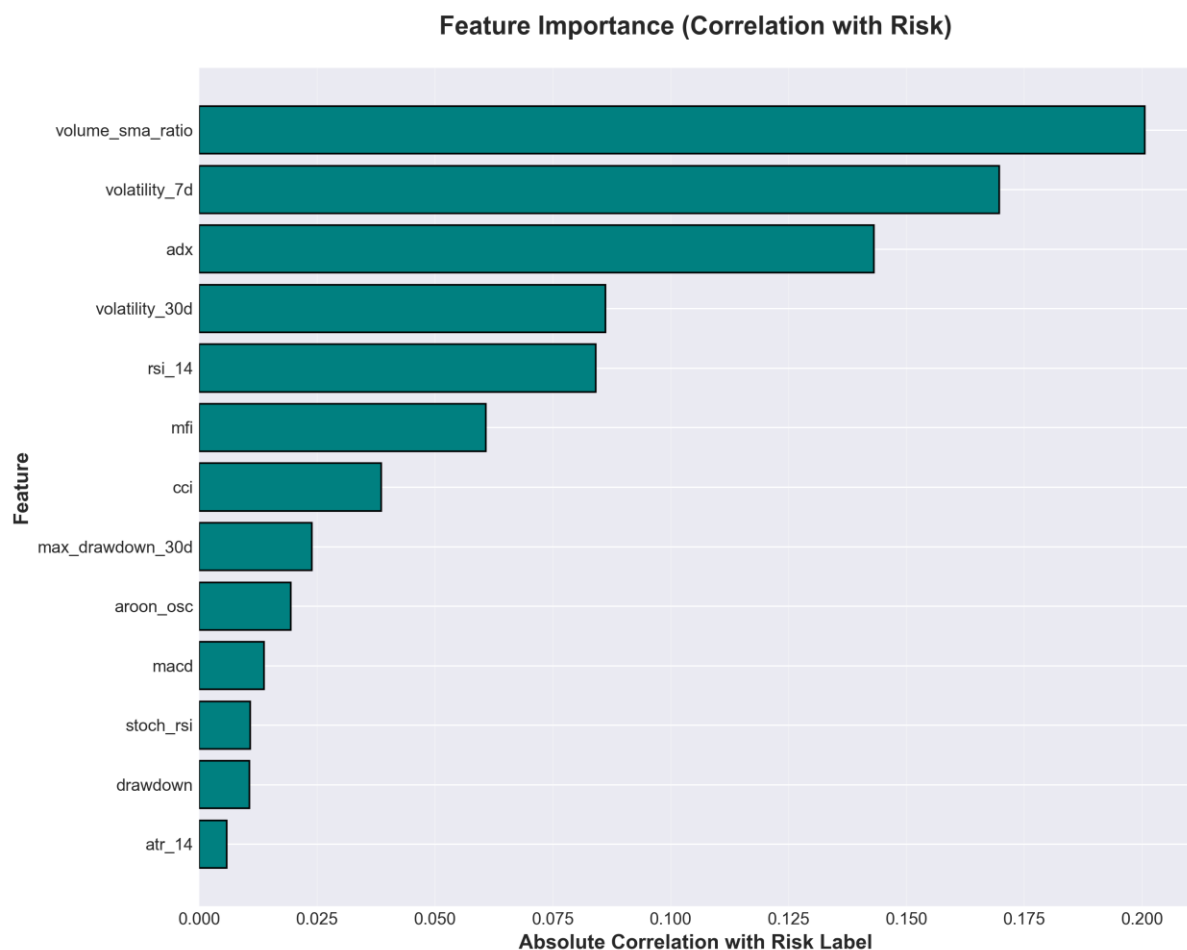


fig 8.2: Feature Importance (Correlation with Risk)

The figure 8.2 highlights the correlation of different features with the risk level, ranked from highest to lowest importance. The most influential feature is volume_sma_ratio, indicating that changes in trading volume relative to its average have the strongest relationship with risk. This shows that unusual volume activity is a key indicator of market instability and high-risk conditions.

Short-term volatility (7-day) and trend strength (ADX) also demonstrate strong correlations with risk, suggesting that rapid price fluctuations and strong directional

movements significantly contribute to higher risk levels. Long-term volatility (30-day) and RSI (14) show moderate influence, indicating that they support risk detection but are less impactful compared to short-term indicators. Lower-ranked features such as MFI, CCI, and drawdown-based metrics exhibit weaker correlations, implying limited contribution to risk prediction. Indicators like MACD, stochastic RSI, and ATR show very low importance, suggesting minimal impact on the model. Overall, the results indicate that volume spikes and short-term market behavior play a more critical role in risk prediction than traditional lagging indicators.

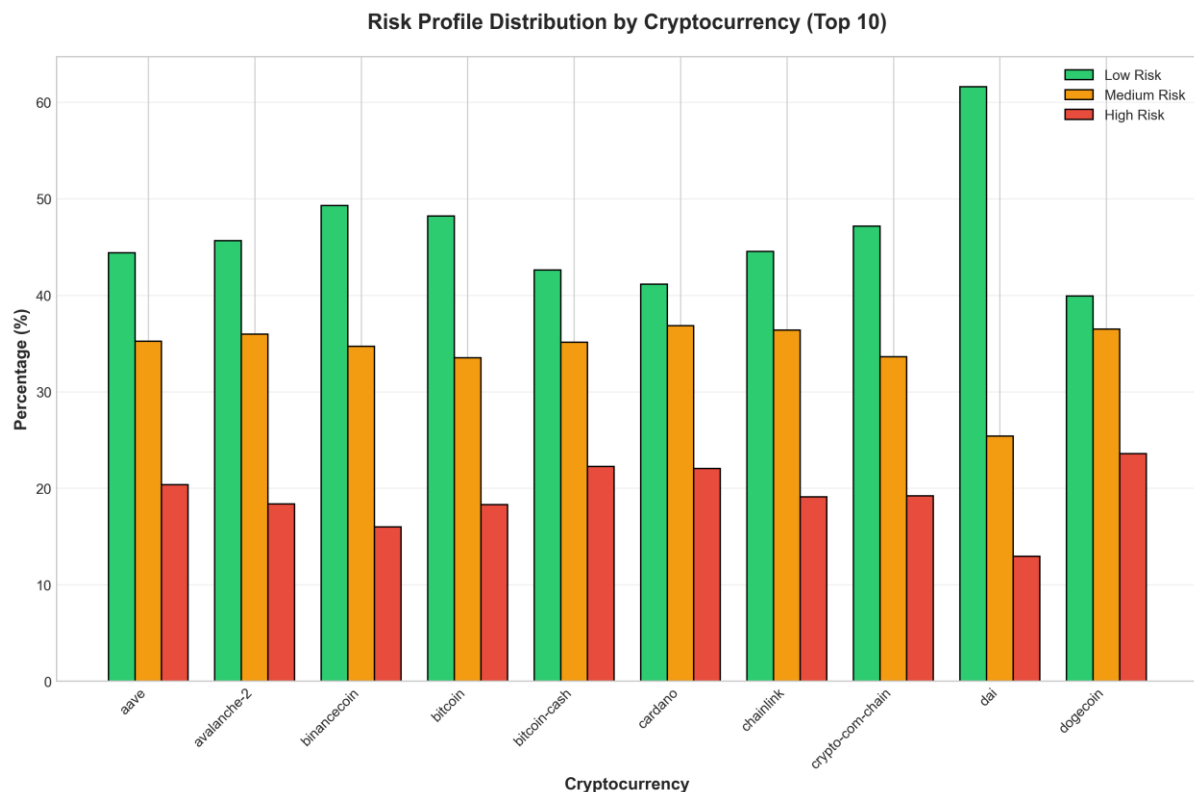


fig 8.3: Risk Profile Distribution by Top 10 Cryptocurrency

The figure 8.3 illustrates the distribution of risk across the top 10 cryptocurrencies, categorized into low, medium, and high risk levels. Most cryptocurrencies are predominantly in the low-risk category, typically ranging between 40% to 60%, indicating that these assets remain in relatively stable or less risky conditions for a significant portion of time.

Certain cryptocurrencies, such as DAI, exhibit a very high percentage of low-risk conditions (above 60%) with minimal high-risk exposure, reflecting its nature as a stablecoin with low volatility. In contrast, assets like Dogecoin, Cardano, and Bitcoin Cash show comparatively higher proportions of high-risk conditions (around 20% or more), suggesting that they experience unstable or volatile market behavior more frequently. The medium-risk

category remains relatively consistent across all cryptocurrencies, generally ranging between 33% to 37%, acting as a transitional phase between stable and high-risk conditions. Overall, the figure indicates that while most cryptocurrencies spend a majority of time in low-risk states, certain assets are more prone to high-risk behavior due to higher volatility and speculative market dynamics.



fig 8.4: Trend Analysis : Risk Metrics across Market Regime

The figure 8.4 compares key risk-related metrics - volatility, drawdown, and RSI across market regimes (0, 1, and 2) for low, medium, and high risk categories. The volatility (7-day) increases from regime 0 to regime 2 for all risk levels, with high-risk conditions showing the highest fluctuations, indicating greater instability in later regimes.

The drawdown analysis shows losses (negative values), where low and medium risk improve gradually across regimes, while high-risk behavior remains inconsistent, indicating unpredictable downside risk. The RSI analysis shows that regime 1 has the highest momentum across all risk levels, while regime 2 shows a decline, suggesting weakening market strength.

Overall, regime 2 represents higher volatility with lower momentum, while regime 1 shows stronger momentum with moderate risk.

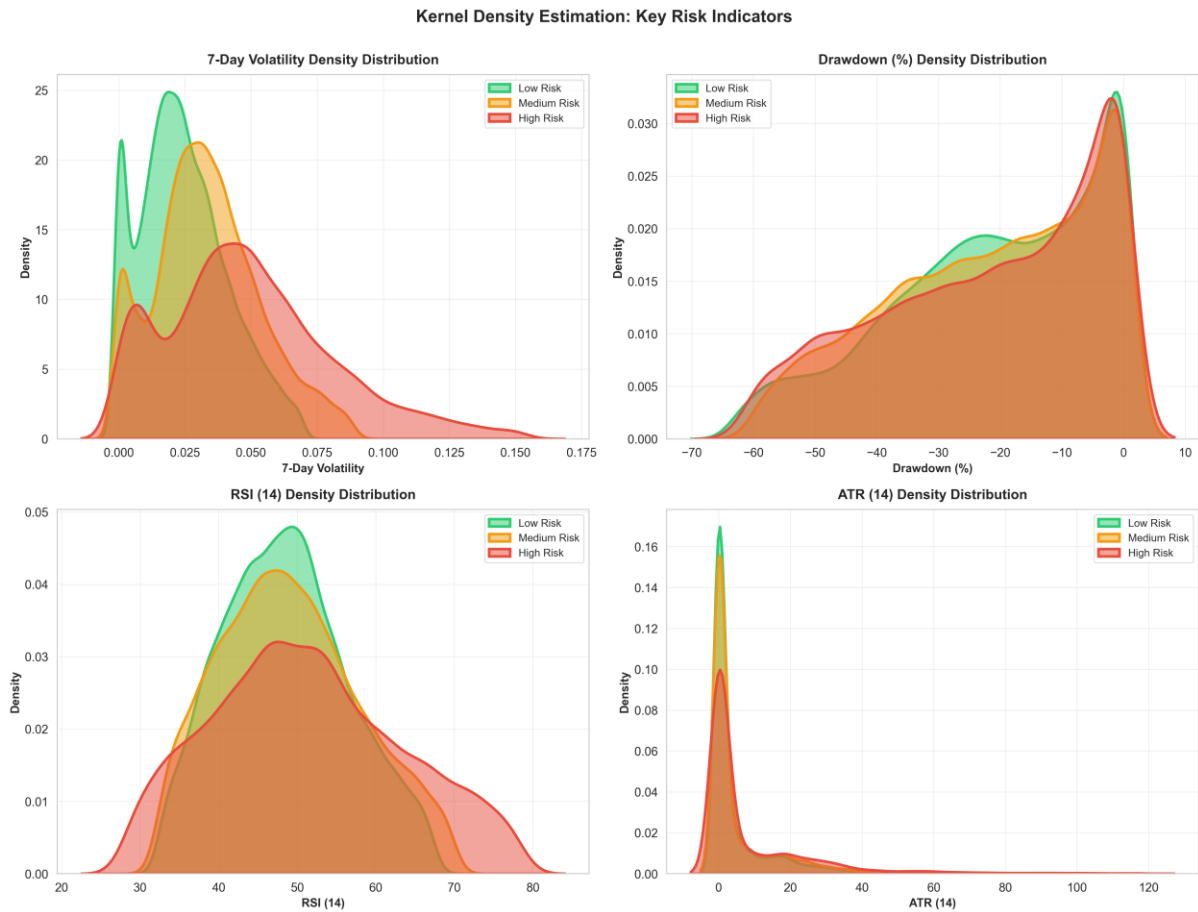


fig 8.5: Kernel Density Estimation : Key Risk Indicators

The figure 8.5 shows the distribution of key risk indicators-volatility, drawdown, RSI, and ATR-across low, medium, and high risk conditions using density curves. The volatility (7-day) distribution shifts significantly to higher values for high-risk conditions, indicating more frequent and extreme price fluctuations, while low-risk values remain concentrated at lower levels.

In the drawdown plot, all categories overlap, but high-risk conditions extend further into deeper losses, showing higher downside risk. The RSI distribution shows low-risk values clustered around the mid-range, while high-risk conditions spread widely, reflecting unstable momentum. Similarly, the ATR plot shows low-risk values concentrated at low ranges, while high-risk conditions extend toward higher values, indicating increased volatility.

Overall, the figure demonstrates that as risk increases, indicator values become more dispersed and shift toward more extreme levels, especially for volatility-related features.



fig 8.6: Model Performance Comparison: Weighted F1 Score

The figure 8.6 shows the performance comparison of five machine learning models-Ensemble, XGBoost (Selected), Gradient Boosting, Random Forest, and Logistic Regression-based on their Weighted F1 Score, with a target threshold of 0.6 indicated by a dashed line as a benchmark for acceptable performance.

Among the models, XGBoost achieves the highest score (0.652) and is highlighted as the selected model, indicating its superior performance. The Ensemble model follows closely with a score of 0.650, while Gradient Boosting (0.640) and Random Forest (0.620) also perform well, exceeding the threshold. In contrast, Logistic Regression records the lowest score (0.570) and fails to meet the required benchmark.

Overall, the figure demonstrates that ensemble and boosting-based models provide better predictive performance compared to simpler models. XGBoost, in particular, offers the most effective balance between precision and recall, making it the most suitable choice for accurate and reliable cryptocurrency risk prediction.

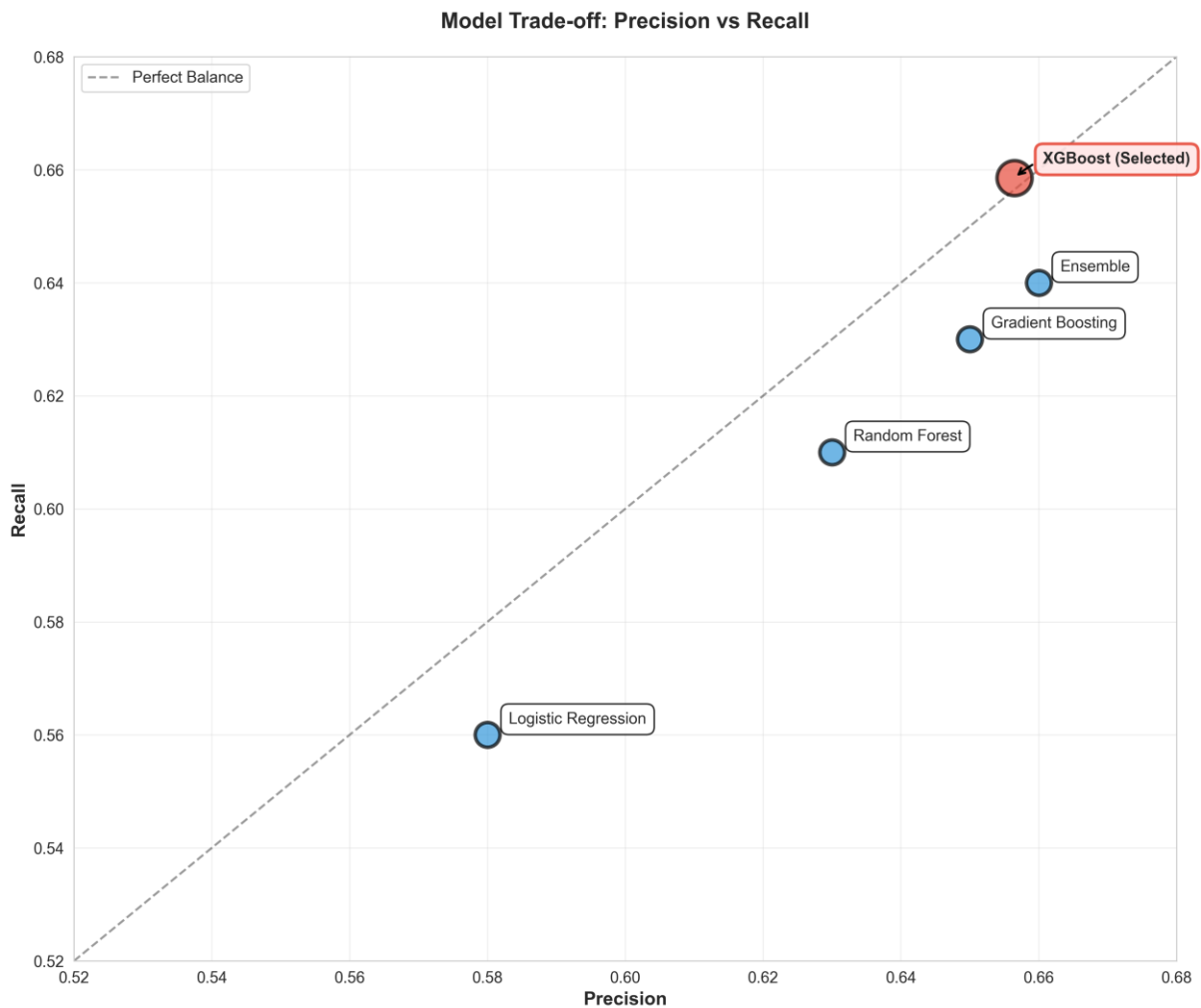


fig 8.7: Model Trade – off Precision vs Recall

The figure 8.7 illustrates the trade-off between precision and recall for different machine learning models, including XGBoost, Ensemble, Gradient Boosting, Random Forest, and Logistic Regression. The dashed diagonal line represents the ideal balance where precision equals recall.

XGBoost (Selected) is positioned closest to this optimal line, achieving high values for both precision and recall, indicating a well-balanced and reliable model. Ensemble and Gradient Boosting also perform well, showing strong precision with slightly lower recall compared to XGBoost. Random Forest demonstrates moderate performance, while Logistic Regression lies further from the ideal line, indicating comparatively lower effectiveness.

Overall, the figure highlights that XGBoost provides the best balance between precision and recall, making it the most suitable model for accurate and consistent risk prediction, while simpler models like Logistic Regression underperform in comparison.

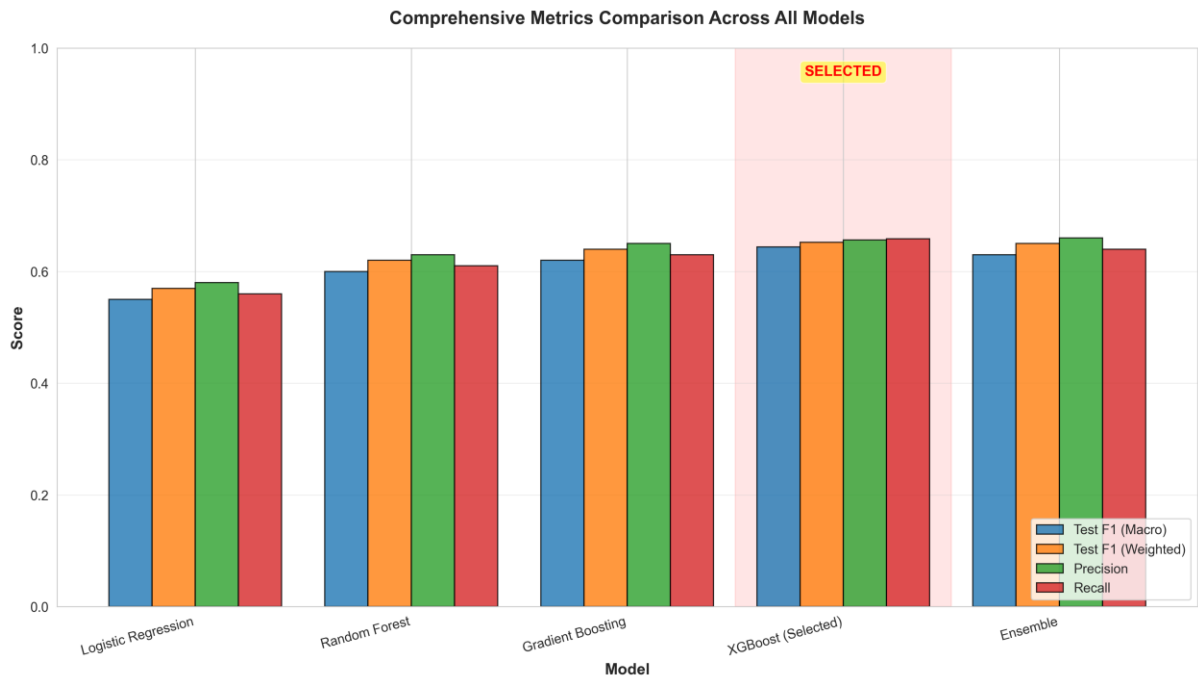


fig 8.8: Comprehensive Metrics Comparison Across All Models

The figure 8.8 presents a comprehensive comparison of multiple machine learning models - Logistic Regression, Random Forest, Gradient Boosting, XGBoost (Selected), and Ensemble based on key evaluation metrics including Test F1 (Macro), Test F1 (Weighted), Precision, and Recall.

XGBoost (Selected) demonstrates the highest performance across all metrics, consistently achieving superior scores in precision, recall, and F1 measures, and is highlighted as the chosen model. Gradient Boosting and Ensemble models also perform strongly, showing competitive results slightly below XGBoost, while Random Forest provides moderate performance. Logistic Regression records the lowest scores across all metrics, indicating comparatively weaker predictive capability.

Overall, the figure clearly shows that advanced boosting and ensemble techniques outperform traditional models, with XGBoost offering the most balanced and reliable performance for accurate cryptocurrency risk prediction.

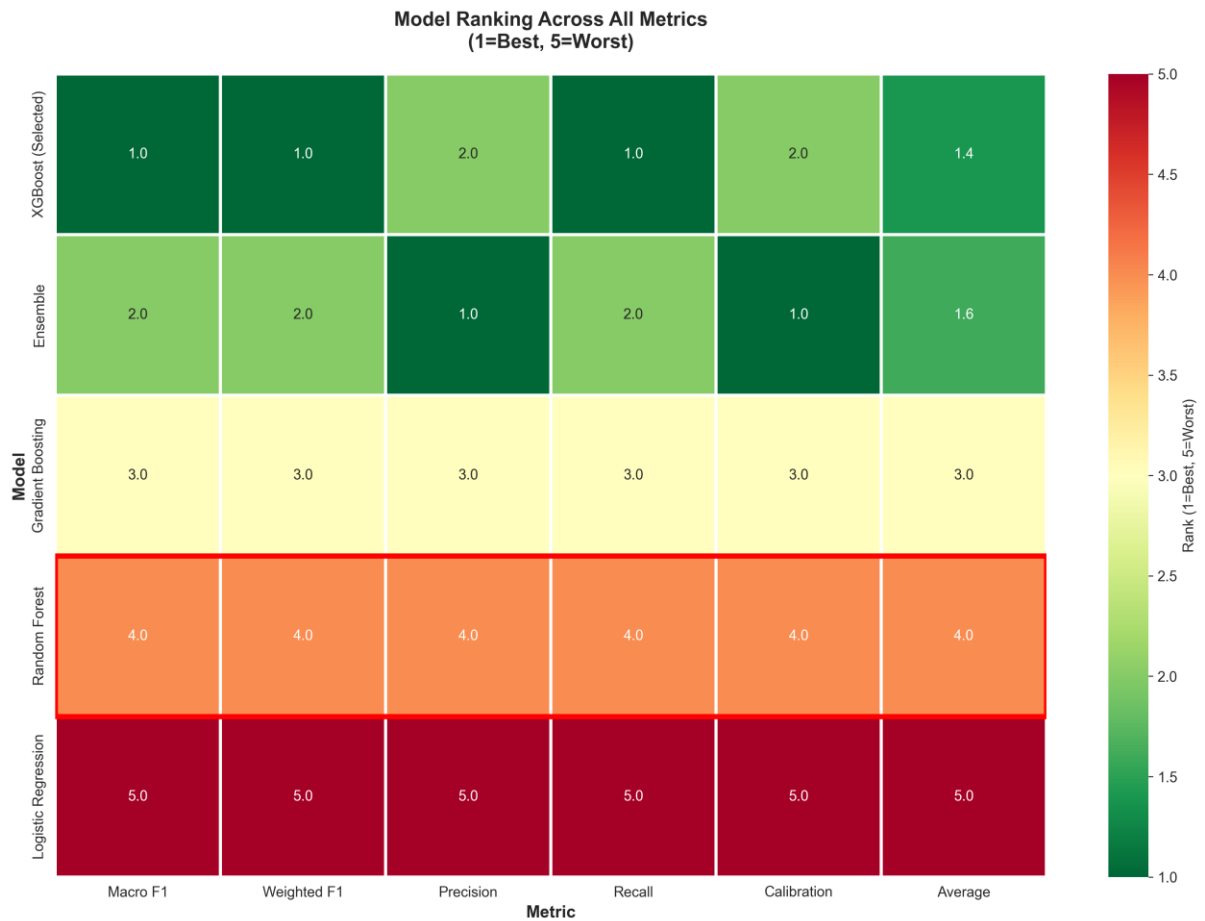


fig 8.9: Model Ranking Across All Metrics (1=Best, 5=Worst)

The figure 8.9 presents the ranking of different machine learning models across multiple evaluation metrics, including Macro F1, Weighted F1, Precision, Recall, Calibration, and overall average performance. The ranking is based on a scale where 1 indicates the best performance and 5 indicates the worst.

XGBoost (Selected) consistently achieves top ranks across most metrics, demonstrating superior and well-balanced performance. The Ensemble model follows closely, securing strong rankings and showing competitive results across all metrics. Gradient Boosting maintains moderate performance with consistent mid-level rankings, while Random Forest ranks lower across all metrics, indicating comparatively weaker performance. Logistic Regression consistently ranks last, reflecting its limited effectiveness for this task.

Overall, the figure clearly highlights that XGBoost is the most reliable and best-performing model, followed by ensemble-based approaches, while simpler models such as Logistic Regression are less suitable for accurate cryptocurrency risk prediction.

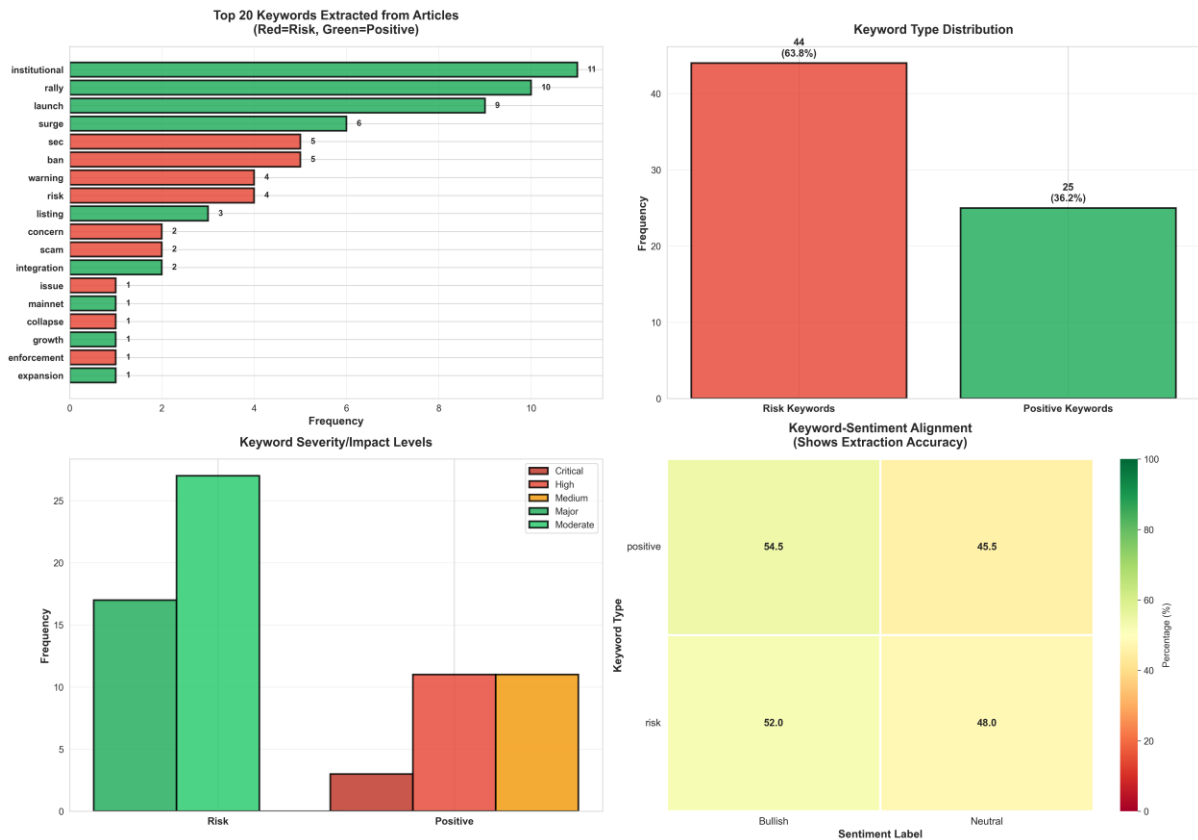


fig 8.10: Sentiment Analysis

The figure 8.10 presents a keyword-based sentiment analysis of cryptocurrency and financial news articles. The analysis shows that risk-related keywords dominate the dataset, accounting for approximately 63.8%, while positive keywords make up around 36.2%. Frequently occurring positive terms include “institutional,” “rally,” and “launch,” whereas risk-related terms such as “SEC,” “ban,” “warning,” and “scam” appear consistently across the dataset.

In terms of severity, most risk keywords fall under the “Major” category, indicating significant negative impact, while positive keywords are distributed between “Critical” and “Medium” levels. The heatmap further reveals a weak alignment between extracted keywords and overall sentiment classification, as both risk and positive keywords appear almost equally in bullish and neutral articles (approximately 52% vs. 48%).

Overall, the figure highlights that while keyword extraction captures important terms, it is not fully reliable for accurately determining sentiment direction, suggesting the need for more advanced sentiment analysis techniques.

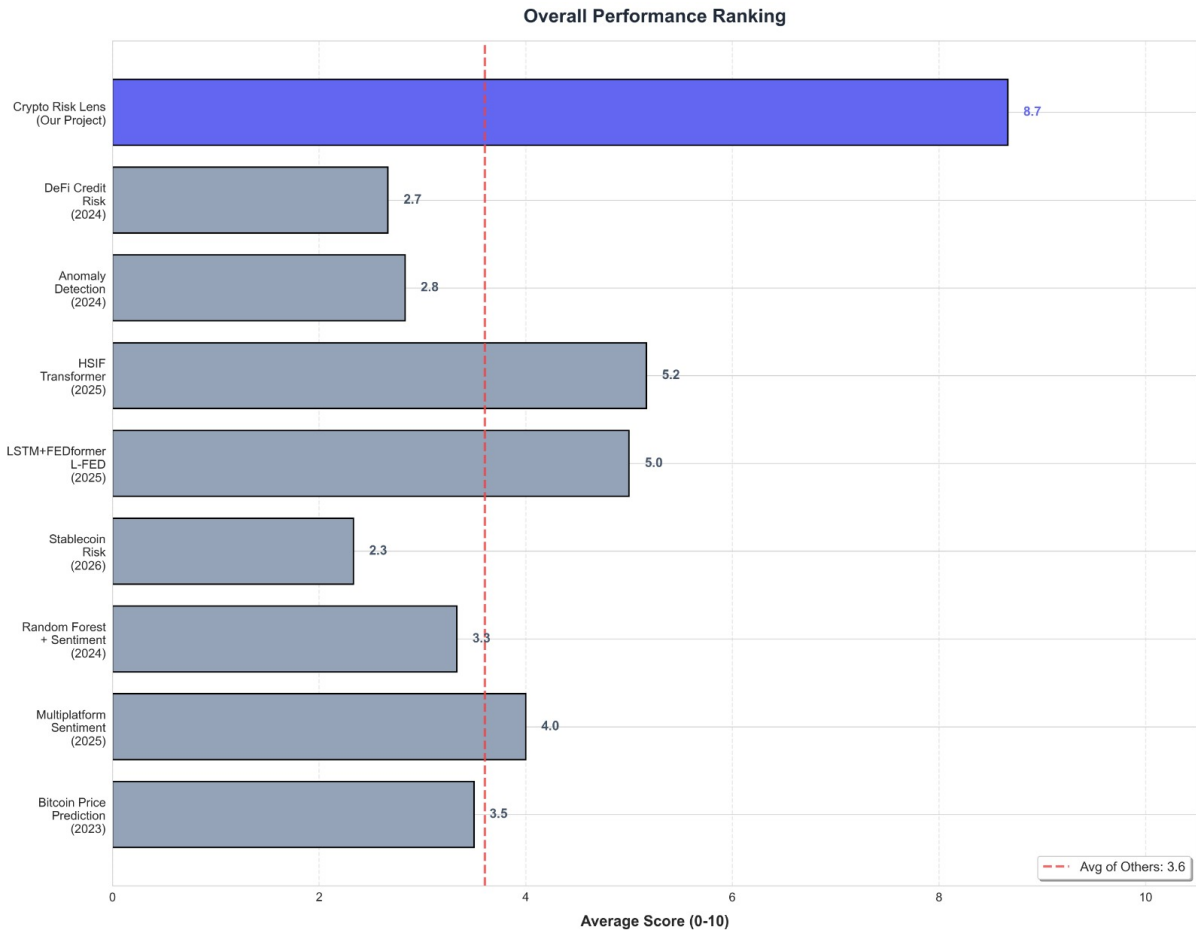


fig 8.11: Overall Performance Ranking

The figure 8.11 presents the overall performance ranking of Crypto Risk Lens compared to existing research approaches based on average scores. Crypto Risk Lens significantly outperforms all other methods with a score of 8.7, clearly exceeding the average benchmark (3.6), which is indicated by the dashed line. This large gap highlights the superior effectiveness of the proposed system in handling complex cryptocurrency risk analysis tasks.

Other approaches such as HSIF Transformer (5.2) and LSTM+FEDformer (5.0) show moderate performance, indicating partial effectiveness but limited overall capability. In contrast, models like DeFi Credit Risk, Anomaly Detection, Stablecoin Risk, and Random Forest with Sentiment score considerably lower, reflecting their inability to capture multiple dimensions of market behavior effectively.

Overall, the figure demonstrates that Crypto Risk Lens provides a substantial improvement over existing methods by combining advanced machine learning techniques, real-time data processing, and comprehensive feature integration. This highlights its

robustness, scalability, and practical applicability for accurate cryptocurrency risk prediction and decision-making.

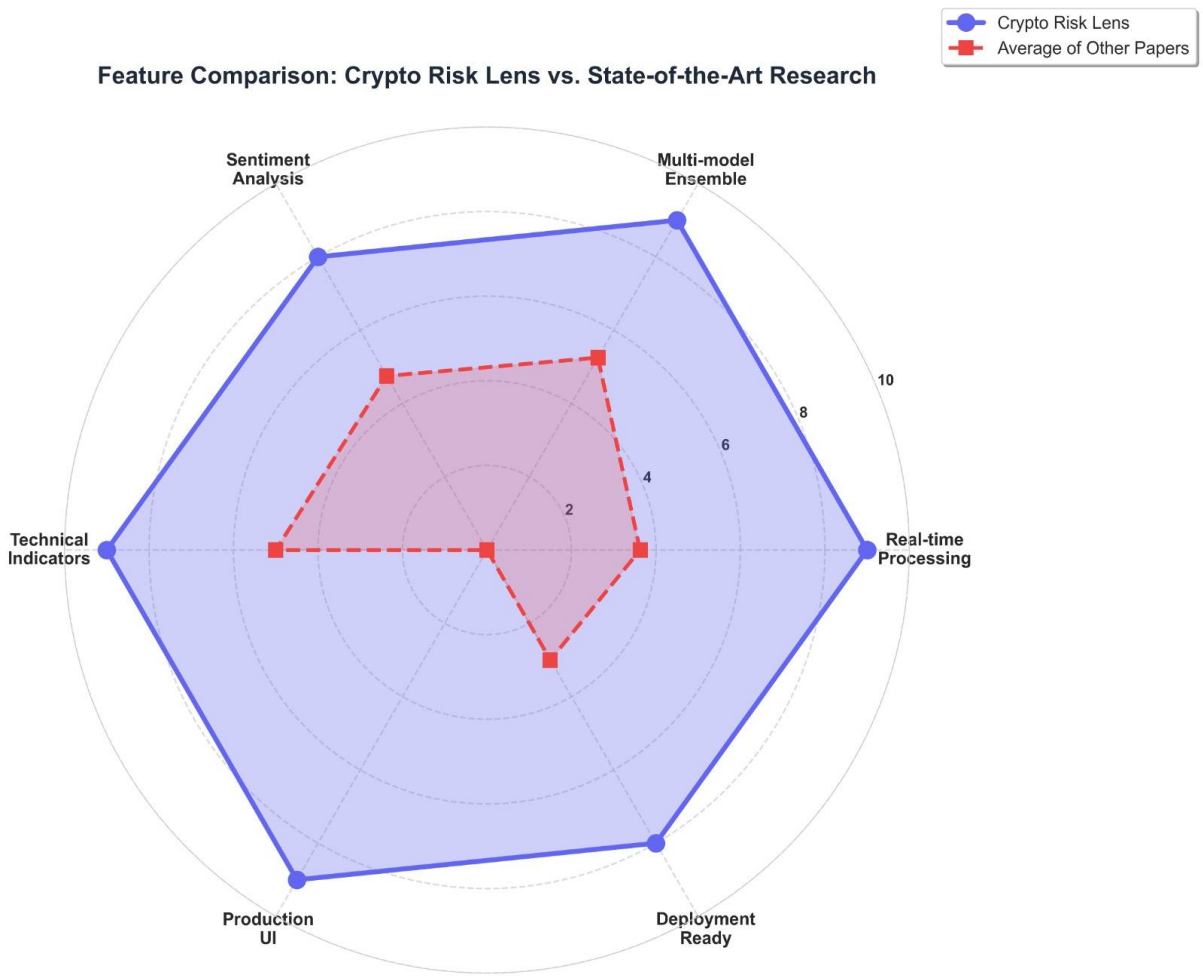


fig 8.12: Feature Comparison: Crypto Risk Lens vs State-of-the-Art Research

The figure 8.12 compares the features of Crypto Risk Lens with state-of-the-art research across key aspects such as real-time processing, multi-model ensemble, sentiment analysis, technical indicators, production UI, and deployment readiness.

Crypto Risk Lens consistently achieves high scores across all categories, particularly excelling in real-time processing, technical indicators, and production-ready user interface. In contrast, existing research approaches show significantly lower scores, especially in deployment readiness and interactive UI capabilities.

Chapter 9

Conclusion

The project was developed with the primary aim of delivering an efficient, reliable, and user-friendly system for real-time cryptocurrency risk analysis. Through systematic requirement analysis, system design, and implementation, the project successfully achieves its objectives by providing a comprehensive platform that integrates machine learning, sentiment analysis, volatility forecasting, and interactive visualization. The system streamlines the process of understanding market behavior and risk, reducing reliance on fragmented tools and enabling more informed and data-driven decision-making.

In addition to its practical utility, the project demonstrates how artificial intelligence and data science techniques can be effectively applied to complex financial domains. The integration of multiple algorithms such as Random Forest, XGBoost, K-Means clustering, Hidden Markov Models, and regression techniques highlights the strength of combining diverse analytical approaches into a unified system. Furthermore, the inclusion of Natural Language Processing for sentiment analysis emphasizes the importance of external factors like public perception in influencing cryptocurrency markets.

The development process provided valuable hands-on experience in areas such as machine learning, backend and frontend development, real-time data processing, and data visualization. Each stage of the project, from data collection and preprocessing to model training, system integration, and deployment, was executed methodically to ensure accuracy, performance, and usability. The iterative approach also reinforced the importance of building user-centric systems that can be easily understood and utilized by both technical and non-technical users.

Overall, the project fulfills its intended goal of enhancing the understanding of cryptocurrency market risk and behavior. Beyond its immediate application, the system is scalable and adaptable, allowing future enhancements such as integration with additional financial markets, advanced predictive models, or improved real-time analytics. Thus, Crypto Risk Lens not only serves as a functional analytical tool but also contributes to ongoing innovation in the fields of financial technology, artificial intelligence, and intelligent decision support systems.

Chapter 10

Future Scope

Expanding Crypto Risk Lens with advanced technologies and additional features can significantly enhance its capabilities and practical impact. The following are potential future scope improvements:

1. Cloud Integration:

Implementing cloud-based infrastructure would enable better scalability, remote access, and efficient handling of large volumes of real-time cryptocurrency data. It would also support distributed processing and allow multiple users to access the platform simultaneously with improved performance and reliability.

2. Advanced Analytics and Notifications:

Incorporating advanced analytics such as long-term trend prediction, portfolio risk scoring, and anomaly detection can provide deeper insights into market behavior. Additionally, automated alerts through email, SMS, or push notifications can notify users about sudden market changes, high-risk conditions, or significant sentiment shifts.

3. Security and Access Control:

Enhancing the system with secure authentication mechanisms, role-based access control, and encrypted data storage can protect sensitive user data and system components. This ensures safe access to analytics and prevents misuse of financial insights.

4. Mobile Platform Extension:

Developing a mobile application would improve accessibility by allowing users to monitor cryptocurrency markets on the go. Features such as real-time notifications, personalized dashboards, and user-specific watchlists can enhance user engagement and usability.

5. Probabilistic Likelihood Estimation:

Integrating advanced AI models such as deep learning and transformer-based architectures can improve prediction accuracy and capture complex market patterns. Additionally, extending the system to support other financial markets such as stocks, forex, and commodities would transform it into a comprehensive financial risk analysis platform.

References

- [1] Jutur Manogna, Gogineni Sravan Chowdary, Gogineni Meghana, & Priyanka C. Nair (2024). “*Bitcoin Price Prediction Based on Sentiment Analysis.*” IEEE Conference Proceedings. [Online]. Available: <https://ieeexplore.ieee.org/document/10440877/>
- [2] Andrew Coutinho & Reshu Agarwal (2024). “*Cryptocurrency Insight: A Multiplatform Sentiment Analysis Framework for Price Prediction.*” IEEE Conference Proceedings. [Online]. Available: <https://ieeexplore.ieee.org/document/11013419/>
- [3] Vincent Oei, Pieter Effendy, Lili Ayu Wulandhari, & Islam Nur Alam (2024). “*Exploring the Effectiveness of Adding Sentiment Analysis into Random Forest to Predict Bitcoin Price.*” IEEE Conference Proceedings. [Online]. Available: <https://ieeexplore.ieee.org/document/10903842/>
- [4] Shalini Singh, Shubhanker Yadav, Anshupriya Tirkey, Beronica Kullu, & Kartikeya Singh (2024). “*Risk Analysis of Stablecoins: Exploring Blockchain Integration and Market Impact.*” IEEE Conference Proceedings. [Online]. Available: <https://ieeexplore.ieee.org/document/10918544/>
- [5] Sumin Li, Rentao Wang, Yudong Wan, & Jincheng Hu (2025). “*Cryptocurrency Price Prediction Using LSTM and FEDformer Enhanced by Sentiment Analysis.*” IEEE Conference Proceedings. [Online]. Available: <https://ieeexplore.ieee.org/document/11299632/>
- [6] Saeed Mohammadi Dashtaki, Mehdi Hosseini Chagahi, Aein Bahadori, Behzad Moshiri, Md. Jalil Piran, & Allahyar Montazeri (2025). “*HSIF: A Transformer-Based Cross-Attention Framework for Cryptocurrency Trend Forecasting via Multimodal Sentiment–Market Fusion.*” IEEE Conference Proceedings. [Online]. Available: <https://ieeexplore.ieee.org/document/10820123/>
- [7] Achraf Yahia, Yassine Mouhssine, Abdelkader El Alaoui, & Said Ouatik Elalaoui (2024). “*Leveraging Machine Learning for Anomaly Detection Methods in Cryptocurrency: A Data-Driven Study.*” IEEE Conference Proceedings. [Online]. Available: <https://ieeexplore.ieee.org/document/10754457/>
- [8] Georgios Palaiokrassas, Sandro Scherrers, Eftychia Makri, & Leandros Tassiulas (2024). “*Machine Learning in DeFi: Credit Risk Assessment and Liquidation Prediction.*” IEEE Conference Proceedings. [Online]. Available: <https://ieeexplore.ieee.org/document/10634435/>